

SOUND SYSTEM DESIGNER

BOXES, HORNS AND WHOLE SYSTEMS

User Guide · Version 1.4.0 · Windows
Swamp Rat Sound · AGPL-3.0-only

CONTENTS

1 Welcome
2 Install and first run
3 A tour of the screen
4 Quick start: your first system
5 The System page
6 The box page
7 Horns and scoops
8 The Build page
9 Home speakers
10 Car audio
11 Brief: design from words
12 Projects and files
13 What the numbers mean, and their limits
14 Troubleshooting
15 Reference tables
16 Glossary

1 WELCOME

Sound System Designer works out loudspeaker enclosures and whole sound systems from driver specifications, and then turns the result into something you can cut: a cut list, baffle drawings, sheet layouts and a bill of materials.

What it does

- **Boxes and horns.** Sealed, ported, band-pass and horn-loaded boxes. Horns can be front-loaded or rear-loaded (a scoop), with exponential, hyperbolic, tractrix or conical flares. Five plots: response, cone excursion, impedance, impedance phase and group delay.
- **Systems.** Up to four bands behind Linkwitz-Riley crossovers, each band a full box or horn design. Trim, delay and polarity per band, multiple drivers and wiring, an amplifier per band, and the summed response in dB SPL with load, power and excursion limits.
- **Home speakers.** A second mode for one cabinet with several drivers, desktop to floorstanding: 2-way, MTM (D'Appolito), WTW, TMM, 3-way and 2.5-way, with dome, planar and woofer classes from 0.75 in to 21 in, a baffle step from the baffle width, a crossover between the drivers, and eight starting points. It builds a one-cabinet cut list with every driver placed on the baffle.
- **Car audio.** A third mode for a car install: one to four subs (single or dual voice coil, series or parallel) in one shared box or a wedge, an amplifier with its load and limits (optionally bridged), the cabin gain of the vehicle, a vehicle space with a fit check, the 12 V side (supply current, cable, fuse, alternator), front speakers (a component set or a coaxial in a door, kick panel or the dash, with adapter rings), the whole system in one plot, and nine starting points.

- **Build.** Wall material and sheet size, box proportions and bracing; a cut list, baffle drawings, sheet layouts, a bill of materials, a 3D preview, SVG and DXF files, a print layout for PDF. Millimetres or inches.
- **Brief.** Describe what you want in words; a language model running on your own computer (Ollama) proposes three variants. It only picks parts and sizes. The same calculators as everywhere else judge the result.
- **Projects.** Save and open `.ssd.json` files; the app also remembers your last project on its own.

What it is not

> **READ THIS FIRST.** Every number is a calculation, not a measurement. The driver classes are invented generic values, not real products, and no brand, driver model or named sound system is modelled. Check anything you build against real drivers and a real amplifier before you trust it, and before you cut wood.

The physics is deterministic: the same inputs always give the same curves. The language model never calculates acoustics; it fills a form that the calculators check.

2 INSTALL AND FIRST RUN

Install

1. Run `Sound System Designer_1.4.0_x64-setup.exe`. It installs for your user only: no administrator rights, a Start menu entry, a desktop shortcut, and an uninstaller in Settings, Apps.
2. The installer is not code-signed, so Windows SmartScreen warns the first time. Choose **More info**, then **Run anyway**.
3. The app needs the Microsoft WebView2 runtime. Windows 11 includes it.
4. Start **Sound System Designer** from the Start menu or the desktop.

Optional: the Brief feature

The Brief panel needs Ollama running on this computer with a model pulled once:

```
ollama pull llama3.1:8b
```

Everything else works without it. The app only ever contacts an Ollama server on `localhost`.

Running in a browser instead

The same app runs in a browser from the source folder: `npm install` once, then `npm run dev` and open the address it prints. In a browser, saving files hands you downloads and opening uses a file picker; in the desktop app you get native Save and Open dialogs.

Uninstalling

Settings, Apps, Sound System Designer, Uninstall. Your saved `.ssd.json` files are yours and are not touched.

3 A TOUR OF THE SCREEN



The header: the page buttons and the two project buttons.

- **PA system / Home speaker.** The switch above the page buttons: a system of boxes (the PA pages), one loudspeaker with several drivers in a cabinet (chapter 9), or a car install (chapter 10). In Home speaker mode the page buttons are Speaker and Build; in Car subwoofer mode Subwoofer and Build. The Brief panel is shown in both and asks for a speaker or a car install.
- **Page buttons.** System, one box button per band (Sub box, Low-mid box, Mid box, High box; with one band it is just Box), and Build.
- **Save project / Open project.** Write and read a `.ssd.json` file. A short message beside them says what happened.
- **Brief.** The panel under the header on every page except Build (chapter 11).
- **About** at the bottom of every page: version, licence and the generic-parts statement.

How the controls behave

- Most numbers are a typed box plus, for the important ones, a slider. Type a value and press Enter or click away. Out-of-range values are pulled back to the nearest allowed value. The allowed ranges are in chapter 15.
- Many sliders are logarithmic (equal steps are equal ratios), which suits frequencies and volumes. Trim, delay and a few others are linear.
- Hovering a plot shows a vertical cursor and a readout at that frequency, shared across all plots on the page.
- **Autosave.** Every change is remembered on this computer, and the app reopens where you left it. Use Save project to keep a copy you can move or back up.

The three layers

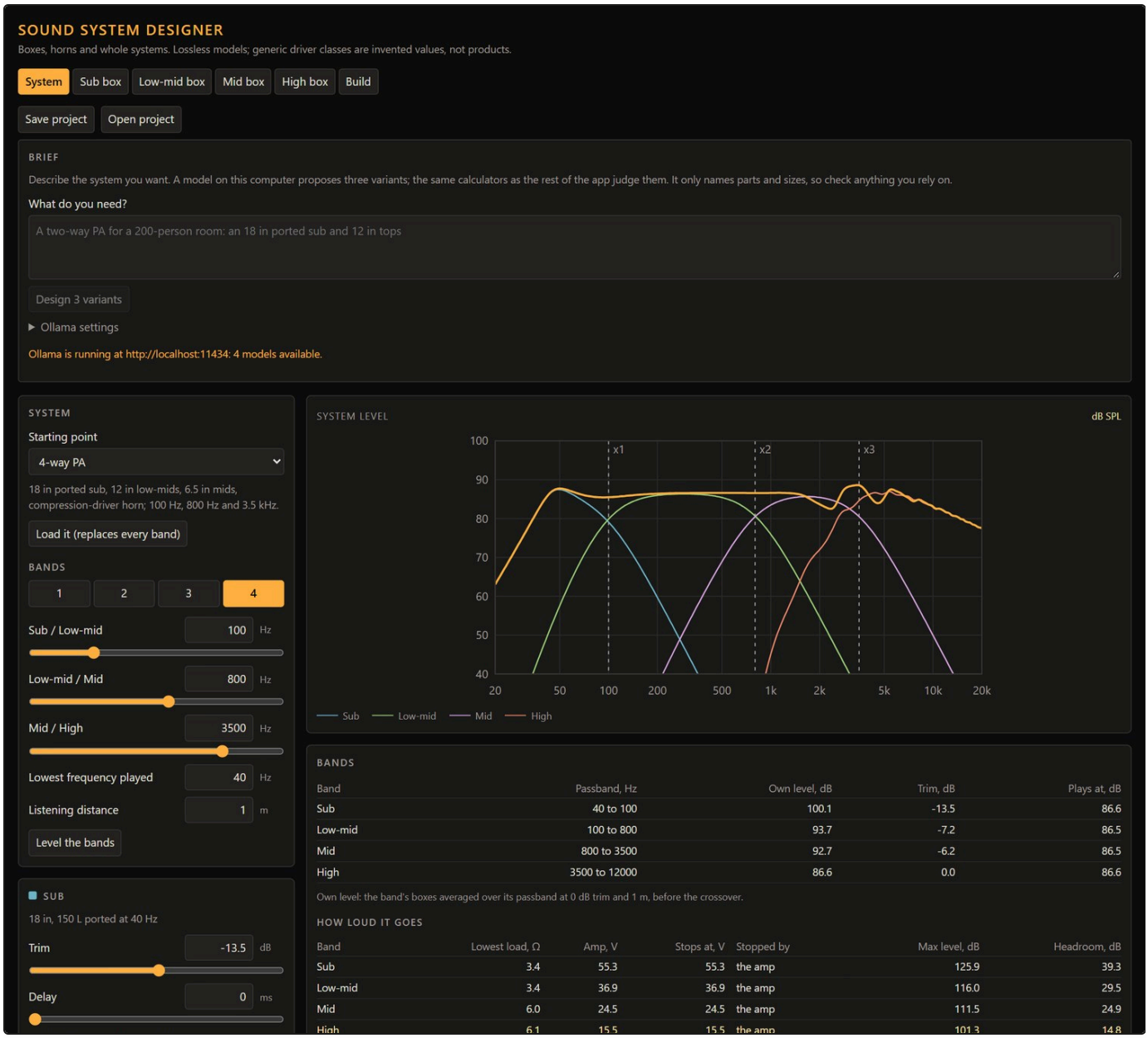
A **project** is a system of 1 to 4 **bands**. Each band is one **box** (a sealed, ported, band-pass or horn design around one driver class) plus its trim, delay, driver count, wiring and amplifier. The **Build** page turns all of it into parts.

4 QUICK START: YOUR FIRST SYSTEM

About five minutes to a cut list.

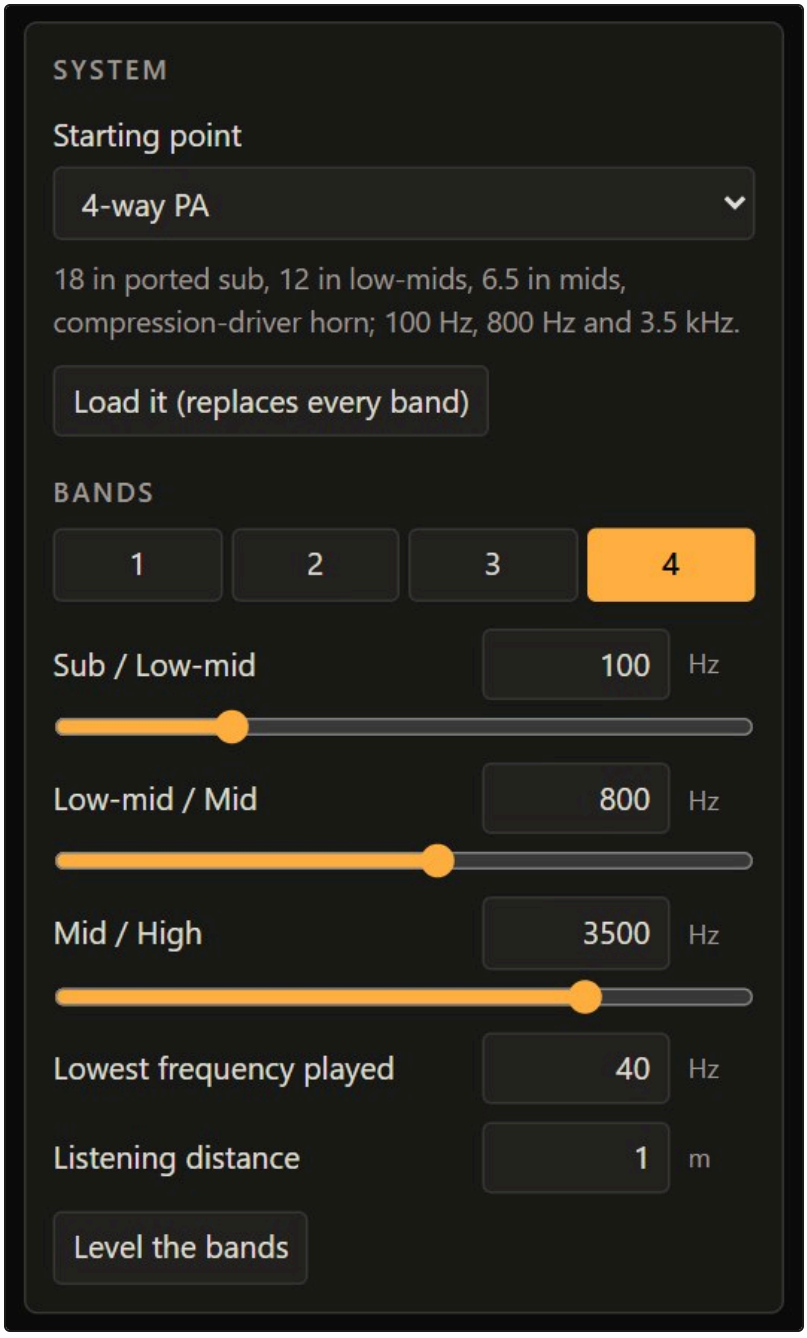
1. On System, open **Starting point**, choose **2-way: tops and sub** and press Load it (replaces every band).
2. Look at the **System level** plot: the orange line is the summed response, the coloured lines are the bands, the dashed vertical line is the crossover. Press Level the bands if one band sits far above the other.
3. Read the **Bands** and **How loud it goes** tables. The caption under the second table names the band that stops first and why.
4. Press Edit this box under the Low band. Change the volume or tuning and watch the response, excursion and the warnings in **Result**.
5. Open Build. Choose the wall material and sheet size, then read the cut list, baffle drawings, sheet layouts and bill of materials.
6. Press Save project to keep it, and Save as CSV file, Download DXF or Export PDF (print) to take it to the shop.

5 THE SYSTEM PAGE



The top of the System page with the 4-way PA starting point: the System section and one section per band on the left, the plot and tables on the right.

The System section



STARTING POINT

Pick one, read the one-line description under it, and press **Load it (replaces every band)**. This replaces every band and crossover; your project name, units and shop settings are kept. Trims are levelled when a starting point loads.

STARTING POINT	WHAT YOU GET
Full range: 12 in sealed	One 12 in sealed box, no crossover.
2-way: tops and sub	18 in ported sub under 12 in sealed tops, crossed at 100 Hz.
3-way PA	18 in ported sub, 12 in sealed mids, compression-driver horn; 100 Hz and 1.5 kHz.
4-way PA	18 in ported sub, 12 in low-mids, 6.5 in mids, compression-driver horn; 100 Hz, 800 Hz and 3.5 kHz.
Jamaican-style: roots dub stack	Four-way: a deep folded 21 in bass bin, 15 in mid bin, 8 in mid horn and a compression-driver horn; 90 Hz, 400 Hz and 2.5 kHz.
Jamaican-style: dancehall stack	Four-way: an 18 in scoop, 12 in mid bin, 6.5 in mid horn and a compression-driver horn; 100 Hz, 900 Hz and 3 kHz.
Jamaican-style: small set	Three-way: a 15 in scoop, a 12 in mid bin and a compression-driver horn; 100 Hz and 2.4 kHz.
Jamaican-style: selector's rig	Two-way: an 18 in scoop and one 8 in horn-loaded top, crossed at 250 Hz.

The Jamaican-style layouts are generic stack types built from generic driver classes and horn geometry. They are not copies of any named sound system.

BANDS AND CROSSOVERS

Press 1 to 4 for the number of bands. Each boundary gets a crossover frequency (20 Hz to 16 kHz) with a slider, named for the two bands it separates. Crossovers are active Linkwitz-Riley fourth-order filters (LR4): the bands sum flat when they are aligned and equally levelled. There is no passive crossover design.

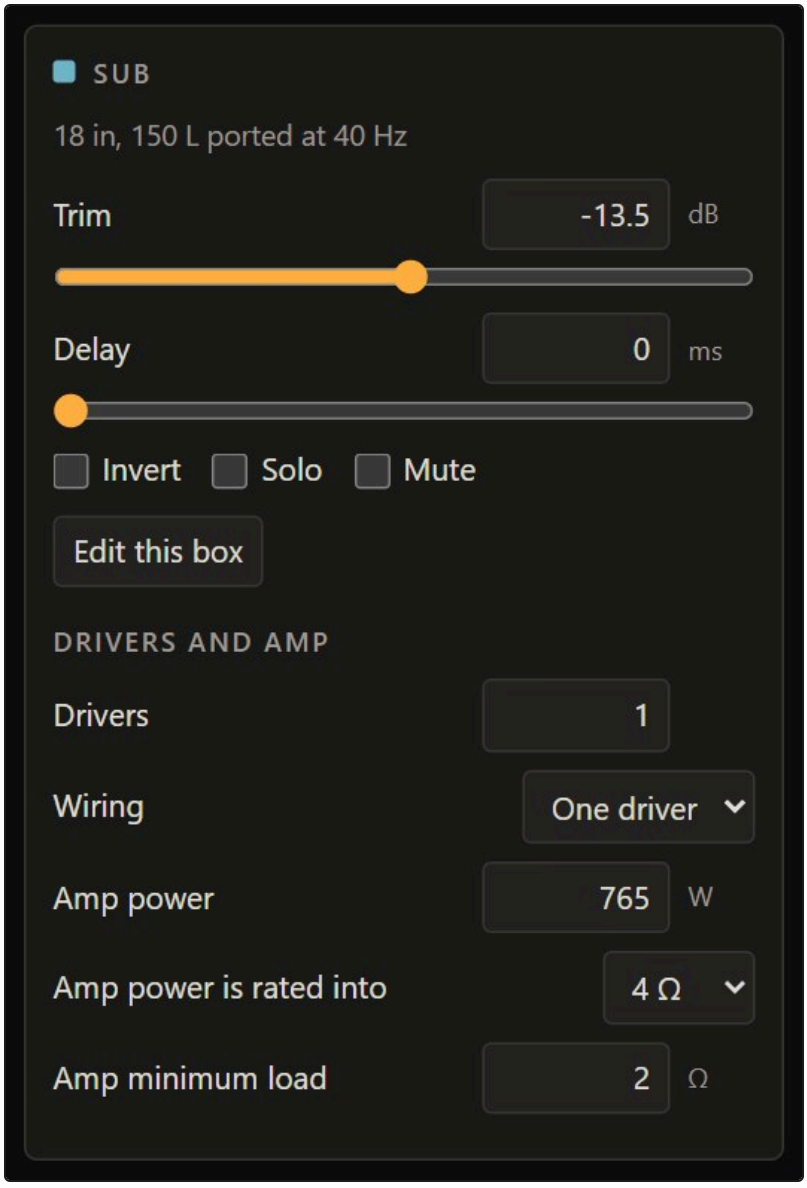
LOWEST FREQUENCY PLAYED, LISTENING DISTANCE

Lowest frequency played (20 to 200 Hz) sets where the average level in the tables starts. **Listening distance** (1 to 200 m) sets the distance for the level estimates; levels are outdoors, half space.

LEVEL THE BANDS

Sets each band's trim so every band plays at the same average level in its own passband. Use it after changing boxes or drivers, then fine-tune by ear and by plot.

A band's section



- The coloured square matches that band's line on the plot. Under the name: driver class, volume and box type.
- **Trim** (-40 to +12 dB) and **Delay** (0 to 20 ms) per band. Delay time-aligns bands whose sources sit at different distances.
- **Invert** flips polarity. **Solo** plays only the soloed band(s). **Mute** silences the band. Muted and non-soloed bands are greyed in the tables.
- Edit this box opens that band's box page (chapter 6).

DRIVERS AND AMP

- **Drivers** (1 to 16): identical drivers in the band, summed as one source. This is optimistic for wide arrays.
- **Wiring:** the options follow the count: all in parallel, all in series, or N in series with M strings in parallel (only splits that divide evenly are offered). Wiring sets the load the amplifier sees.

- **Amp power** (1 to 20000 W), **Amp power is rated into** (a load in ohms) and **Amp minimum load** (1 to 16 ohm).

The plot and tables



BANDS

Band	Passband, Hz	Own level, dB	Trim, dB	Plays at, dB
Sub	40 to 100	100.1	-13.5	86.6
Low-mid	100 to 800	93.7	-7.2	86.5
Mid	800 to 3500	92.7	-6.2	86.5
High	3500 to 12000	86.6	0.0	86.6

Own level: the band's boxes averaged over its passband at 0 dB trim and 1 m, before the crossover.

HOW LOUD IT GOES

Band	Lowest load, Ω	Amp, V	Stops at, V	Stopped by	Max level, dB	Headroom, dB
Sub	3.4	55.3	55.3	the amp	125.9	39.3
Low-mid	3.4	36.9	36.9	the amp	116.0	29.5
Mid	6.0	24.5	24.5	the amp	111.5	24.9
High	6.1	15.5	15.5	the amp	101.3	14.8

The system can be raised 14.8 dB above the reference input before High stops (the amp): about 101 dB SPL average at 1.0 m. Max level is each band's own passband level at 1 m, at the voltage where it stops.

- Levels are for 2.83 V into a band at 0 dB trim, in half space, at 1.0 m outdoors. Horn bands are radiated power, not on-axis level.
- Limits are for a steady sine and ignore power compression and any limiter: real programme has peaks above its average, so treat them as ceilings, not operating points. Each driver is taken to be in its own box.

System level (dB SPL) with the band curves, the Bands table and How loud it goes.

SYSTEM LEVEL

The summed response in dB SPL at the listening distance (orange) with each audible band's own curve in its colour. Dashed lines `x1`, `x2`, `x3` are the crossovers.

BANDS TABLE

COLUMN	MEANING
Passband, Hz	The band's span between its crossovers (the lowest band starts at the lowest frequency played).
Own level, dB	The band's boxes averaged over its passband at 0 dB trim and 1 m, before the crossover.
Trim, dB	The trim you set.
Plays at, dB	Own level plus trim. Bands that play at about the same level sum flat.

HOW LOUD IT GOES

COLUMN	MEANING
Lowest load, Ω	The lowest impedance the amplifier sees in the band's passband, with your wiring.
Amp, V	The most voltage the amplifier can deliver into that load.
Stops at, V	The voltage at which the band hits its first limit.
Stopped by	**the amp** (out of voltage or power), **driver power** (thermal rating) or **cone excursion** (Xmax).
Max level, dB	The band's own passband level at 1 m at the voltage where it stops.
Headroom, dB	How far the band can be raised above the reference input before it stops.

The caption names the limiting band: the whole system can be raised by that band's headroom, to about the stated dB SPL average at your listening distance.

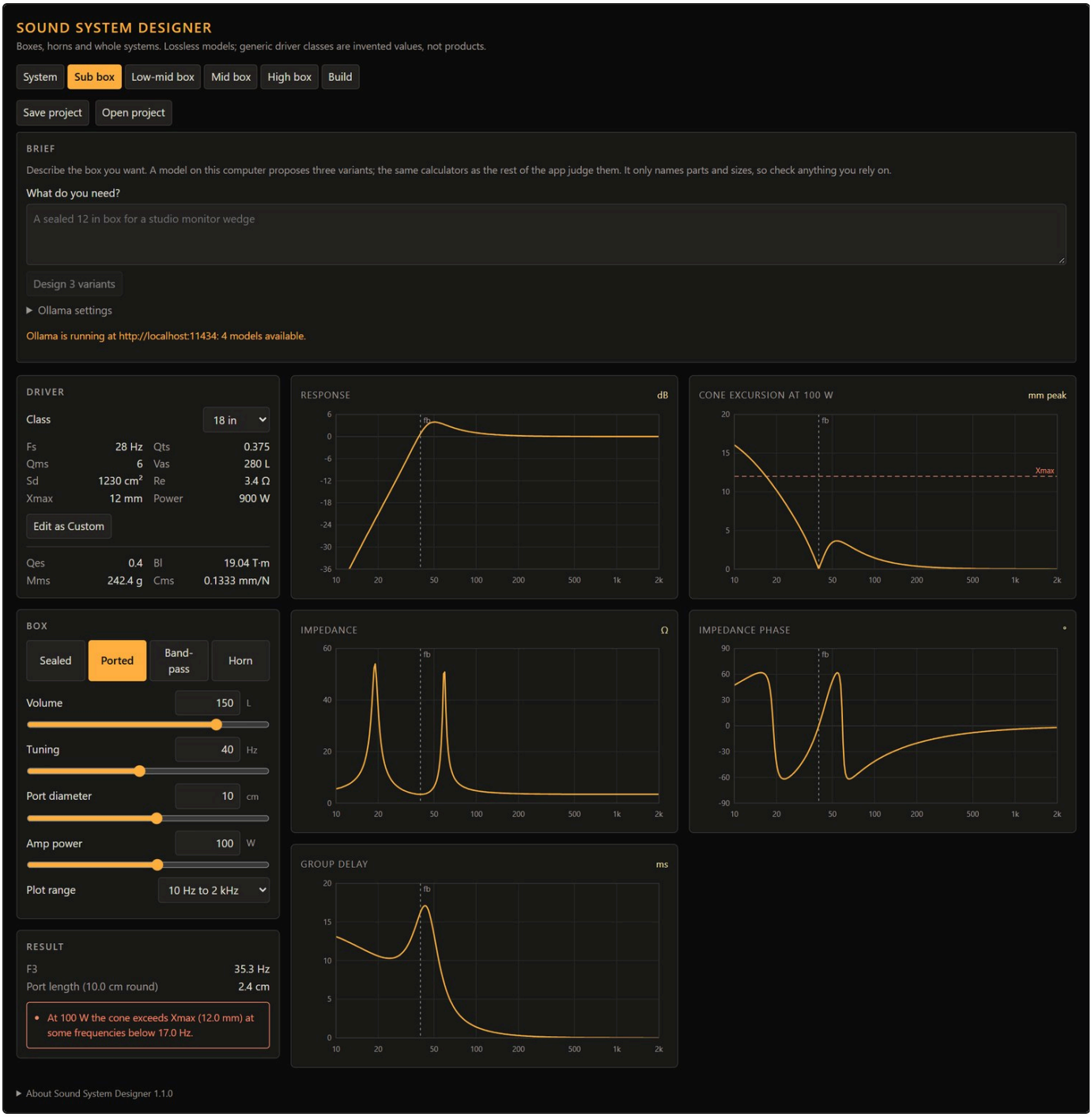
WARNINGS AND NOTES

Red boxed items are warnings (for example a load under the amplifier's minimum, an amplifier that can exceed the driver's power rating, a cone passing Xmax, or two bands cancelling at their crossover). Plain bullets are notes about how the numbers were made. Read both; the warnings are the calculators judging your design.

> **SMALL-SIGNAL.** Limits are for a steady sine and ignore power compression and limiters. Real programme has peaks above its average. Treat Max level as a ceiling, not an operating point.

6 THE BOX PAGE

Open it with **Edit this box** or the band's box button. It designs one band's enclosure around one driver.

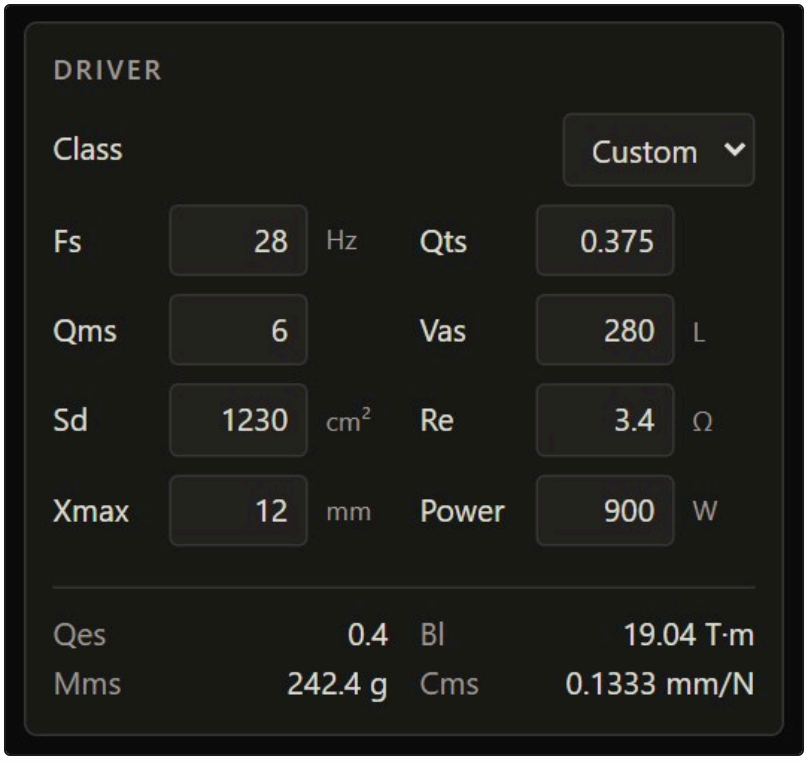


The box page for the Sub band: Driver, Box and Result on the left, five plots on the right.

Driver

Choose a **Class**: eight generic classes (10, 12, 15, 18 and 21 in woofers, 8 in and 6.5 in mid drivers, and a 1 in compression-driver stand-in) or **Custom**. The class's Thiele-Small parameters are shown; the derived values

(Qes, Bl, Mms, Cms) are calculated from them.



Edit as Custom copies a class's values into editable fields.

For a real driver, press **Edit as Custom** (or pick Custom) and enter its datasheet values: Fs, Qts, Qms, Vas, Sd, Re, Xmax and rated Power. The allowed ranges are in chapter 15.

> **THE 1 IN HF CLASS** is a stand-in. Compression drivers rarely publish Thiele-Small values, so read a horn built on it as a sketch.

Box

Four types: **Sealed**, **Ported**, **Band-pass** and **Horn** (chapter 7). The fields change with the type:

FIELD	USED BY	MEANING
Volume	all	Internal air volume (L). For band-pass and horn it is the sealed rear chamber.
Tuning	ported, band-pass	Port resonance, Hz (band-pass: the front chamber and port resonance).
Front volume	band-pass	The ported front chamber (L).
Port diameter	ported, band-pass	Round port, cm. The port length is calculated.
Amp power	all	The power used for the excursion plot (W).
Plot range	all	10 Hz to 2 kHz, or 100 Hz to 20 kHz (for mids and horns).



Sealed: a gentle roll-off, no port.



Band-pass: a narrow passband between two chambers.

Result

The readouts depend on the type: box resonance **fc**, **Qtc** and **F3** (sealed); **F3** and port length (ported; *does not fit* means the port is longer than the box can hold); **peak** and **passband** (band-pass); length, mouth and compression ratio (horn). F3 is where the response is 3 dB below its passband level.

Warnings you may see: a port that does not fit; the cone exceeding Xmax at your chosen power, and from which frequency; power above the driver's rating; Qms below Qts.

The five plots

PLOT	SHOWS	WHAT TO LOOK FOR
Response	dB, relative to the passband	A flat top and a clean roll-off. A hump before the roll-off is a ported box that is tuned too high, or a loose alignment.
Cone excursion	mm peak at the chosen power, with the Xmax line	Stay under Xmax. A ported box's excursion dips to near zero at the tuning frequency and climbs steeply below it.
Impedance	ohms	Peaks at the box resonances. The minimum is what the amplifier must drive.
Impedance phase	degrees	How reactive the load is. Large swings stress some amplifiers.
Group delay	ms	Time smearing near the low cut-off. Ported boxes peak near their tuning.

Vertical dashed lines mark `fb` (port tuning), `fc` (box or flare cutoff). Move the pointer over any plot for a readout; all five follow.

7 HORNS AND SCOOPS

Choose **Horn** as the box type to get the Horn section. The horn model is one-dimensional and lossless: a real transmission-line simulation of the flare, not a rule of thumb.



The Horn page, top part: the Horn section and the plots; the response has the lumped bin overlaid (dashed). The Result and the panel table are further down the page.

Starting points

Four buttons load a complete generic design: **Bass horn**, **Scoop**, **Mid horn** and **HF horn**. They set the driver class, chamber and flare. Start from the nearest one and adjust.

Loading

- **Front-loaded:** the cone fires into the horn throat, with a sealed chamber behind it.
- **Rear-loaded (scoop):** the cone's front radiates directly and its rear drives the horn through the chamber behind it. The cone-to-mouth distance is not modelled; the model adds the two outputs as if they sat in one place.

Flare and geometry

CONTROL	MEANING
Flare	Exponential, Hyperbolic, Tractrix or Conical.
Flare cutoff	The lowest frequency the horn loads (15 Hz to 2 kHz). It sets the horn's length for every flare except conical.
T	Hyperbolic only: 1 is exponential, 0 is catenoidal.
Throat area, Mouth area	cm^2. A mouth smaller than the full size for the cutoff loses bass.
Length	Conical only (m); other flares derive length from the cutoff.
Throat chamber / Chamber behind cone	The air space between the cone and the throat (front-loaded), or behind the cone (scoop). L.
Full-size mouth for the cutoff	A button that sets the mouth to the smallest area that fully loads the chosen cutoff.

Result and the lumped bin

The Result lists length, mouth (and what percentage of full size it is), mouth area, horn air volume and the compression ratio (cone area : throat area). The horn response is **radiated power**, not on-axis pressure, so a horn looks flat up to its upper limit rather than rolling off.

A second, dashed curve on the response overlays the simpler lumped horn-bin model, for comparison.

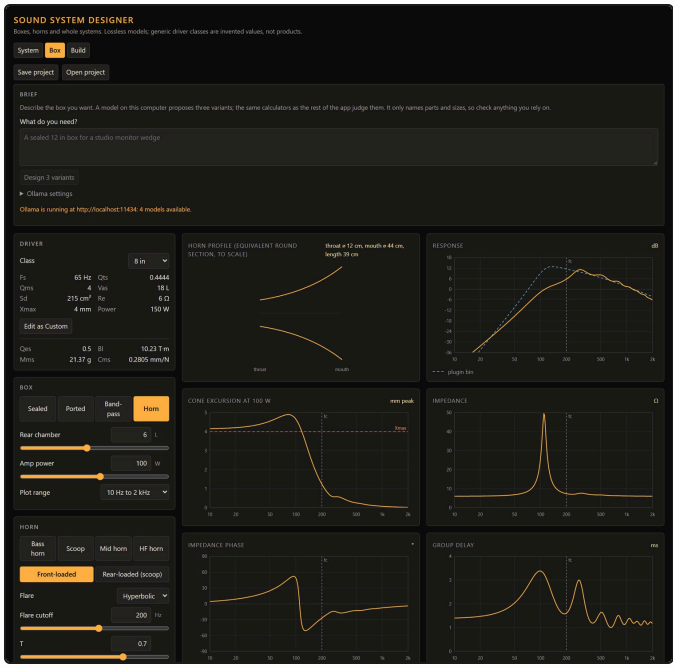
Panels

Real horns are built from straight panels. Type the **straight panel lengths along the centreline** in cm, separated by commas or spaces (for example `40, 40, 50, 60`), and set the **Horn width**. The model then simulates the horn as built, with the ideal flare overlaid, and the table lists the area and height at each panel joint. Left empty, the table shows the ideal flare at eight stations.

> **NO BENDS.** Lengths are along the centreline and there is no bend correction. If your panel lengths do not add up to the flare's length, the last panel is cut short or stretched to end at the mouth, and a warning says so.



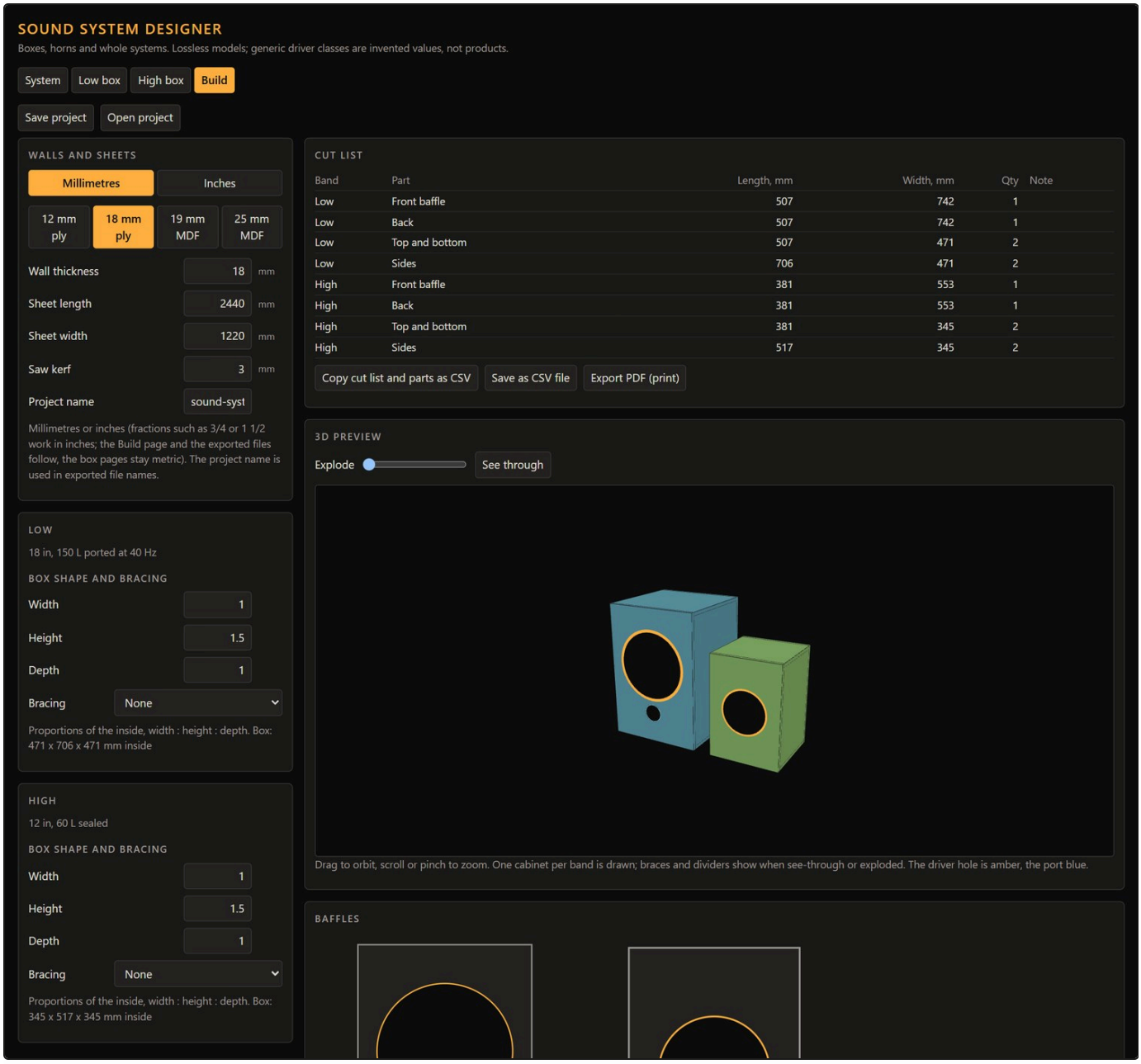
Scoop: rear-loaded bass bin.



Mid horn: a hyperbolic flare with a 200 Hz cutoff.

8 THE BUILD PAGE

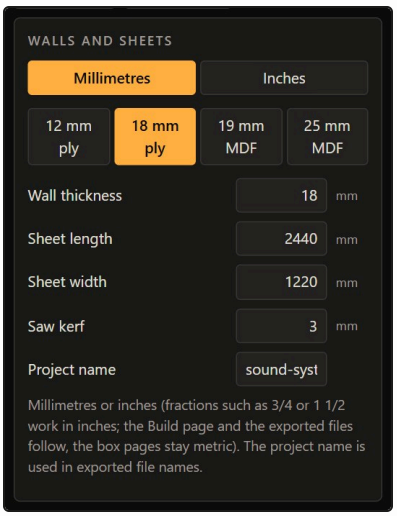
The Build page turns every band into a rectangular box you can cut. It follows the number of bands and the units you choose.



The top of the Build page for the 2-way starting point.

Walls and sheets

- **Units:** Millimetres or Inches. In inches, fractions work: `3/4`, `1 1/2`, `19 mm`. Switching swaps the defaults (3/4 in wall, 48 x 96 in sheet, 1/8 in kerf) unless you changed a value.
- **Wall material:** 12 mm ply, 18 mm ply, 19 mm MDF, 25 mm MDF (in inches, the inch equivalents), or type any thickness (3 to 50 mm).
- **Sheet length / width** and **Saw kerf** (0 to 10 mm).
- **Project name:** used in the names of exported files.

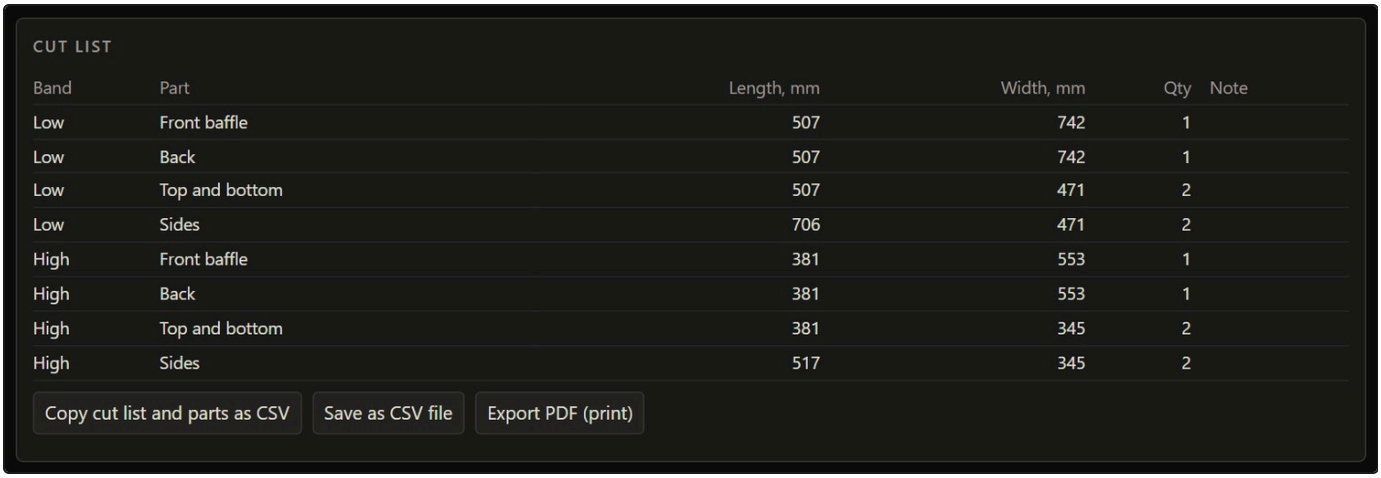


Per band: box shape and bracing

Width : Height : Depth are the *proportions* of the inside (each 0.2 to 5), not sizes. The app finds the dimensions that hold exactly the designed air volume, after subtracting the driver's, the port's and any braces' displacement. **Bracing** is *None*, *One window brace* or *Window braces every 0.4 m*. A small line under the controls states the resulting inside size. Band-pass boxes get a divider and two chambers; horns get a row of sections.



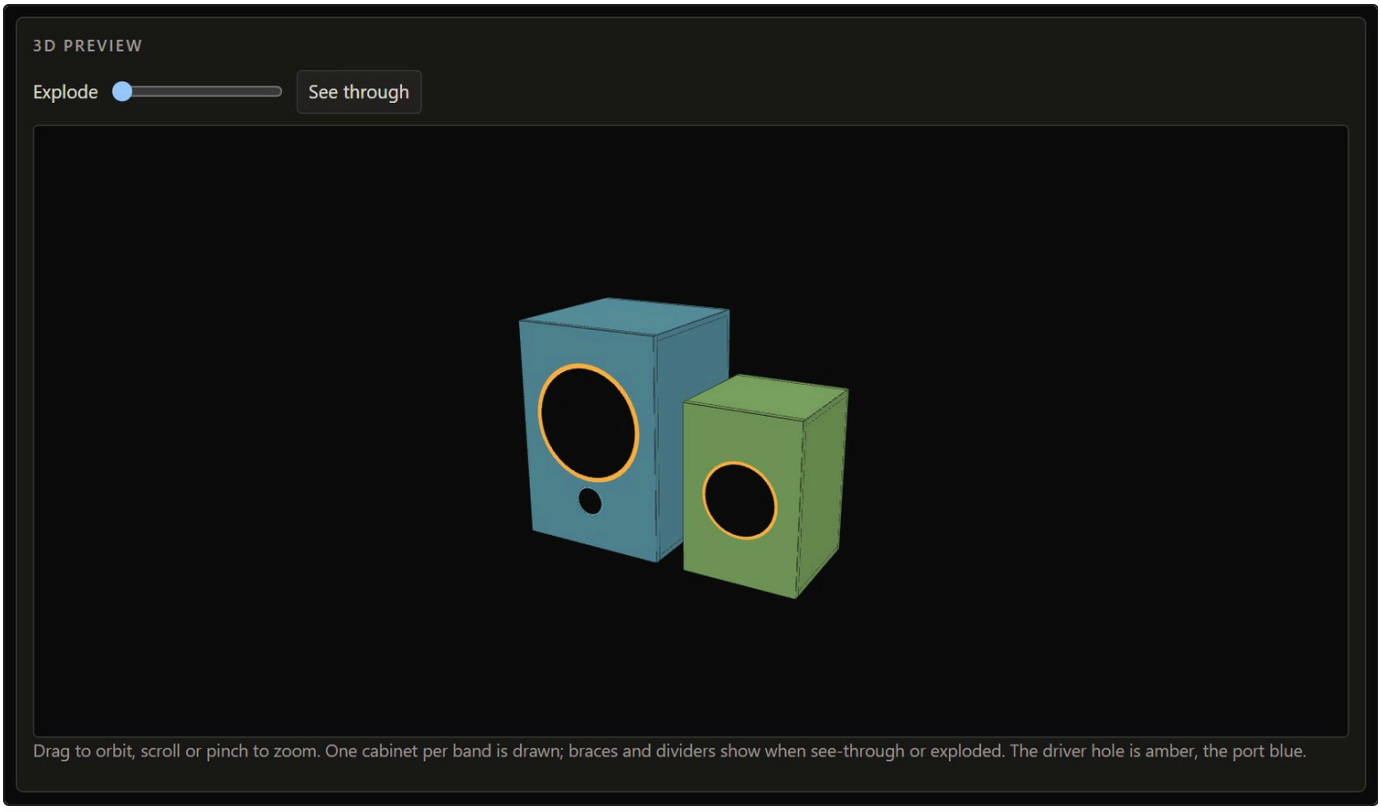
Cut list



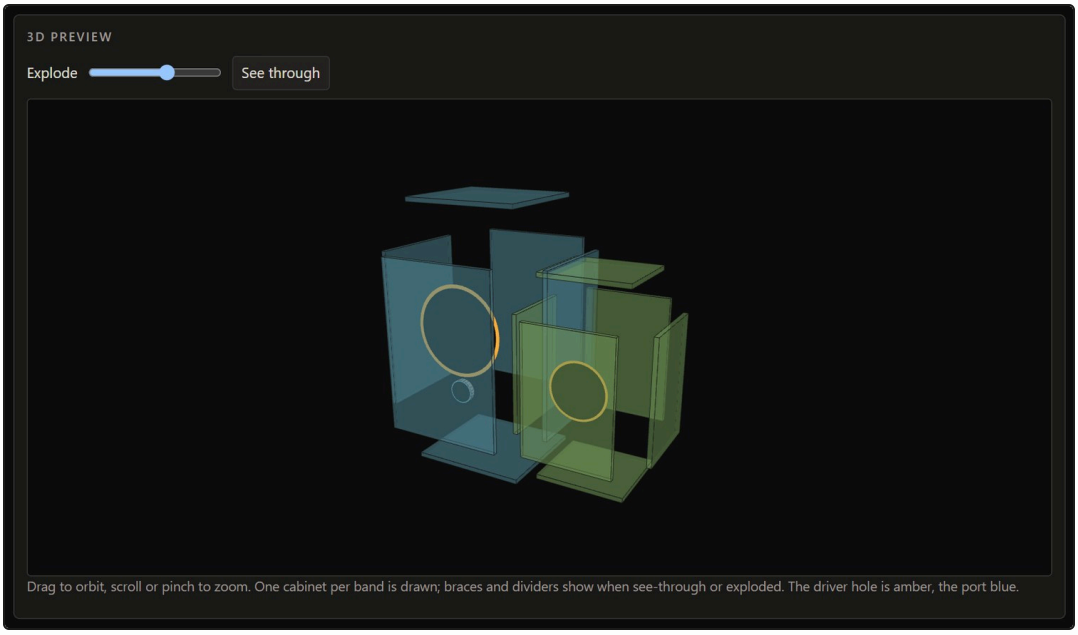
One row per part type: band, part, length, width, quantity and a note. Parts are front baffle, back, top and bottom, and sides (plus braces, dividers and horn sections where the design has them), with butt joints assumed: the sides run between the top and bottom, the baffle and back sit between all four. Three buttons:

- Copy cut list and parts as CSV puts a spreadsheet-ready version on the clipboard.
- Save as CSV file writes it to a file.
- Export PDF (print) opens the print dialog with a clean layout (controls and the 3D view hidden): choose **Save as PDF**.

3D preview

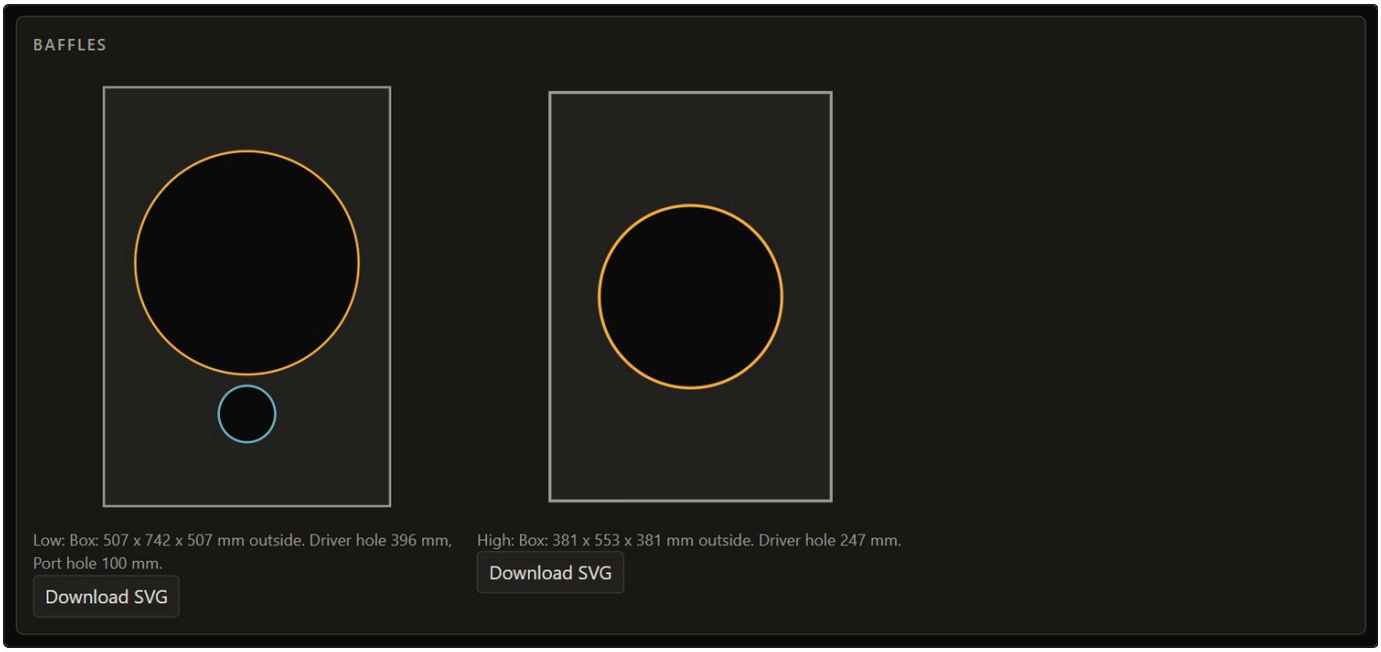


One cabinet per band, drawn from the cut list with the driver hole in amber and the port hole in blue. Drag to orbit; scroll or pinch to zoom. **Explode** pulls the panels apart; See through makes them transparent so braces and dividers show.



Exploded and see-through: the panels, braces and dividers.

Baffles



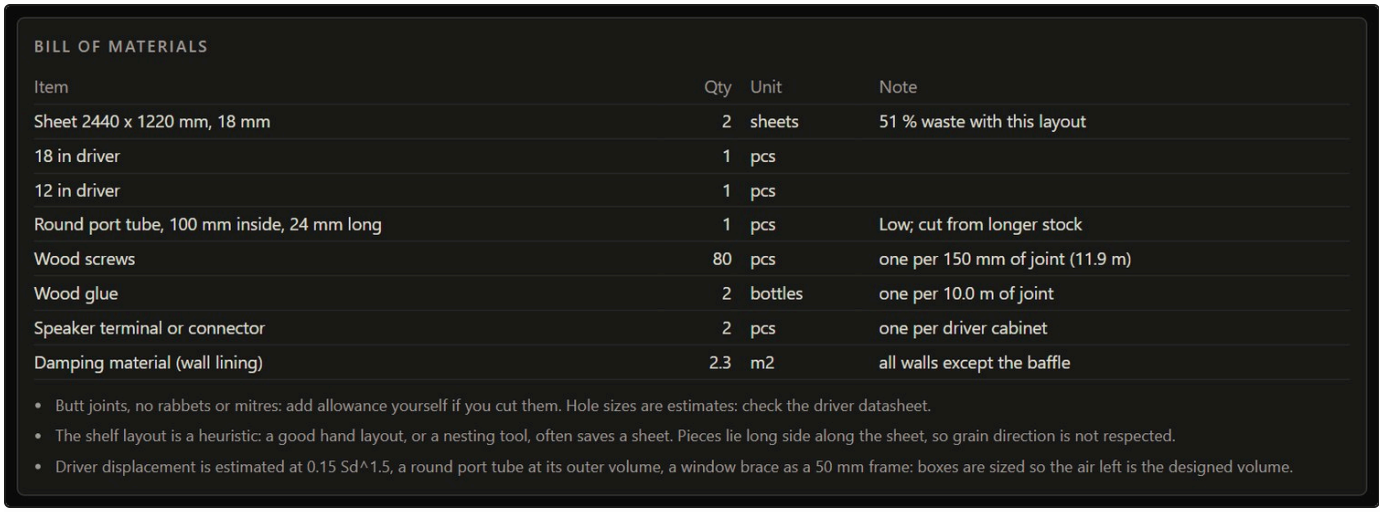
A to-scale front baffle for each cabinet with the driver and port holes. The caption gives the outside size and the hole diameters. Download SVG saves the drawing.

Sheets



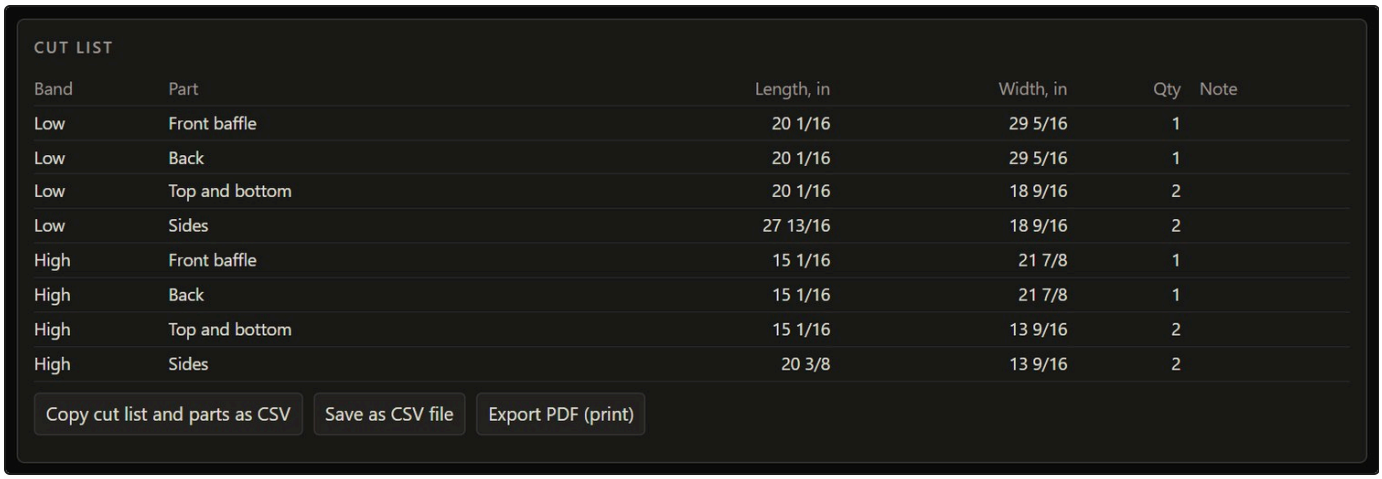
The pieces are laid out on your stock sheets, long side along the sheet, coloured by band. The heading gives the sheet count and the percentage of waste. **Download SVG** and **Download DXF** save each sheet. The DXF is ASCII R12, in millimetres or inches to match your units setting, with layers `SHEET`, `CUT` and `LABELS`, ready for a CNC or laser CAM tool. The layout is a simple heuristic, not an optimal nesting.

Bill of materials and warnings



Sheets, drivers, port tubes, screws, glue, terminals and damping, with quantities and notes. Warnings appear below it: a driver or port hole that does not fit the baffle, a piece larger than a sheet, or a horn approximated because no panel lengths were entered.

Inches



CUT LIST

Band	Part	Length, in	Width, in	Qty	Note
Low	Front baffle	20 1/16	29 5/16	1	
Low	Back	20 1/16	29 5/16	1	
Low	Top and bottom	20 1/16	18 9/16	2	
Low	Sides	27 13/16	18 9/16	2	
High	Front baffle	15 1/16	21 7/8	1	
High	Back	15 1/16	21 7/8	1	
High	Top and bottom	15 1/16	13 9/16	2	
High	Sides	20 3/8	13 9/16	2	

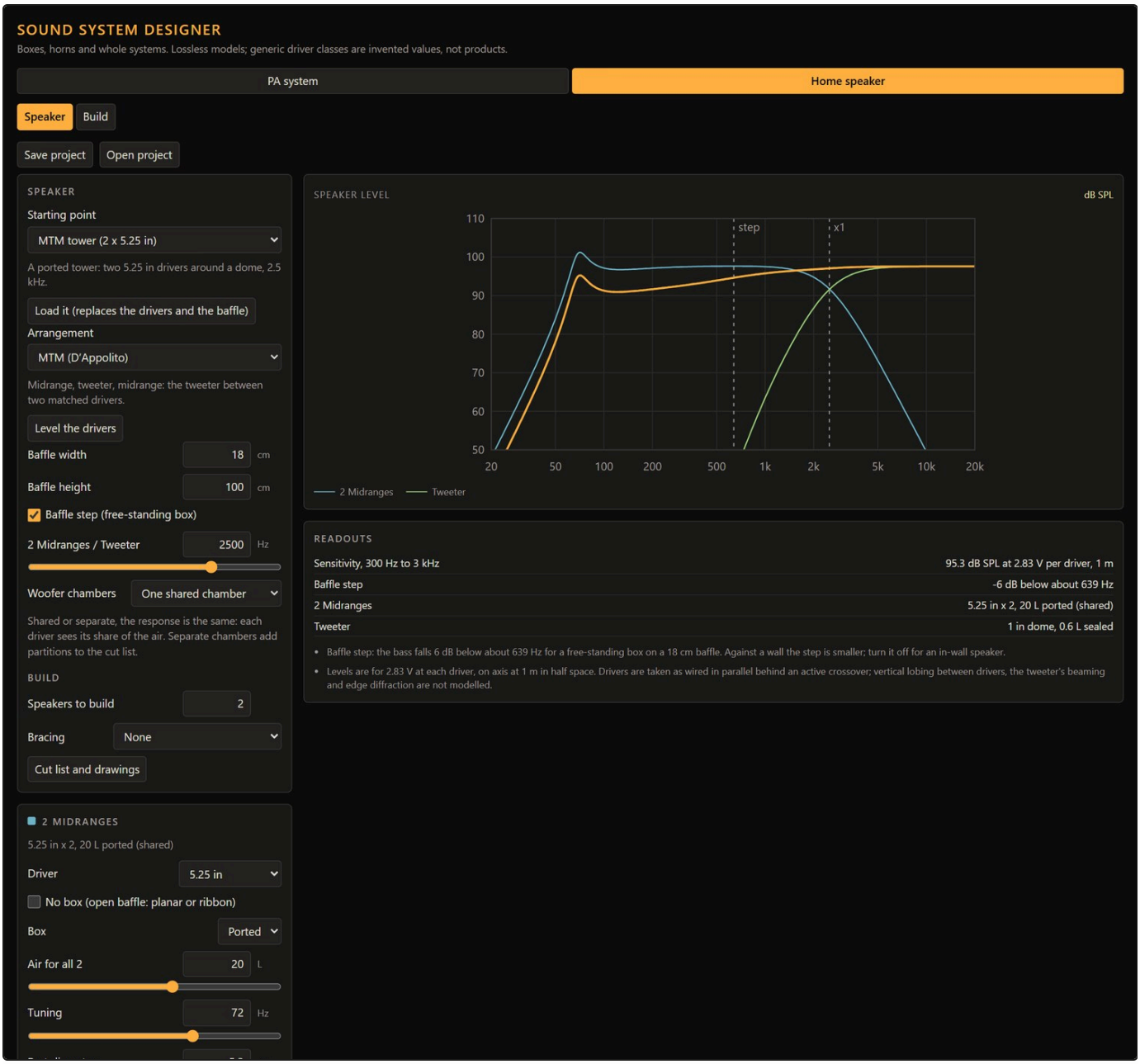
Copy cut list and parts as CSV | Save as CSV file | Export PDF (print)

The same cut list in inches, in shop fractions to 1/16.

In inches, lengths show as fractions to 1/16 in the cut list, captions and bill of materials; SVG is sized in inches and DXF is written in inches. **The box pages stay metric** (litres, cm, cm^2).

9 HOME SPEAKERS

Switch the header to **Home speaker** to design one cabinet that holds several drivers: a tweeter and a woofer, a tweeter between two midranges, a three-way tower. The PA system pages stay as they are; both designs live in the same project and you can switch back and forth.



The Speaker page with the MTM tower starting point.

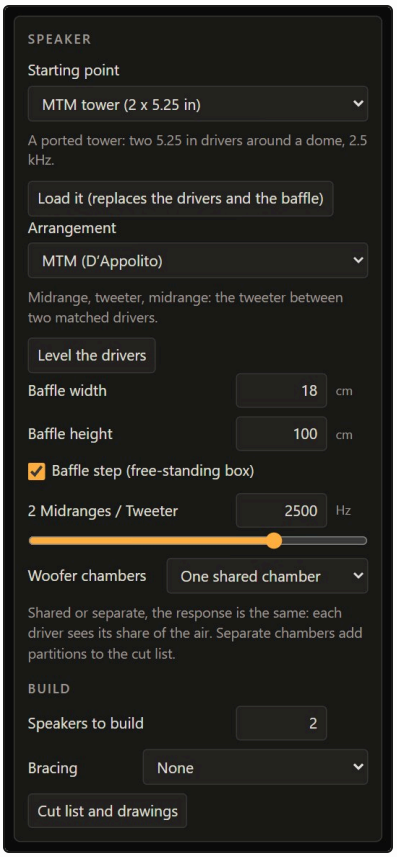
Starting points

Choose one under **Starting point** and press **Load it (replaces the drivers and the baffle)**. Loading keeps how many speakers you are building, the bracing and the chamber setting, and levels the drivers.

STARTING POINT	DRIVERS	BAFFLE, CM	CROSSOVER
Desktop 2-way (4 in)	4 in woofer, 3 L sealed; 0.75 in dome	14 x 22	3 kHz
Bookshelf 2-way, sealed (5.25 in)	5.25 in woofer, 8 L sealed; 1 in dome	20 x 32	2.5 kHz
Bookshelf 2-way, ported (5.25 in)	5.25 in woofer, 12 L ported; 1 in dome	20 x 36	2.5 kHz
Floorstander 2-way (8 in)	8 in woofer, 25 L ported; 1.25 in dome	24 x 95	1.8 kHz
D'Appolito bookshelf (2 x 4 in)	two 4 in mids, 5 L sealed; 1 in dome	16 x 40	2.5 kHz
MTM tower (2 x 5.25 in)	two 5.25 in, 20 L ported; 1 in dome	18 x 100	2.5 kHz
3-way floorstander	10 in woofer, 40 L ported; 4 in mid; 1 in dome	28 x 110	350 Hz, 3 kHz
2.5-way tower (2 x 5.25 in)	two 5.25 in woofers, 8 L each; 1 in dome	20 x 85	300 Hz, 2.5 kHz

These are generic, invented parts, not real products. They are places to start; change anything.

The Speaker section

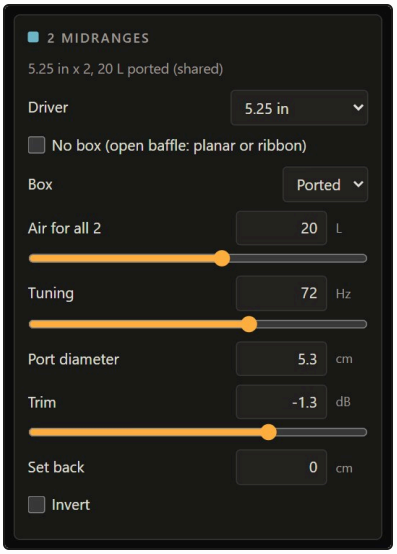


- **Arrangement.** *2-way*; *MTM (D'Appolito)*, a dome between two matched drivers; *WTW*, the same layout with woofers; *TMM*, the dome on top of two drivers; *3-way*; and *2.5-way*, two woofers of which the second plays only the bass. Changing the arrangement keeps a driver whose role is unchanged (a woofer stays a woofer) and gives a fresh driver to any slot whose role changed, so a tweeter never becomes a midrange. The baffle and the crossovers restart at the new arrangement's.
- **Baffle width and height.** The outside size of the front panel, in centimetres. The width sets the baffle step.
- **Baffle step.** A free-standing box loses 6 dB of bass below about 115 / width (in metres): the speaker radiates into a full sphere there and into half a sphere above. Leave it on for a box on a stand; turn it off for an in-wall speaker.
- **Crossover sliders** between the drivers (100 Hz to 8 kHz). Points closer than 1.2 times are moved apart and the page says so. In a 2.5-way the first slider is where the second woofer stops, not a crossing.

- **Level the drivers.** Trims each driver so its own level over the range it plays matches the quietest one's, which stays at 0 dB. Pressing it twice changes nothing more.
- **Woofer chambers** (for pairs): one shared chamber or one per driver. The response is the same either way, because each driver sees its share of the air; separate chambers add a horizontal shelf between them to the cut list and the 3D view.
- **Speakers to build** (1 to 8; 2 is a stereo pair) and **Bracing** for the cut list.

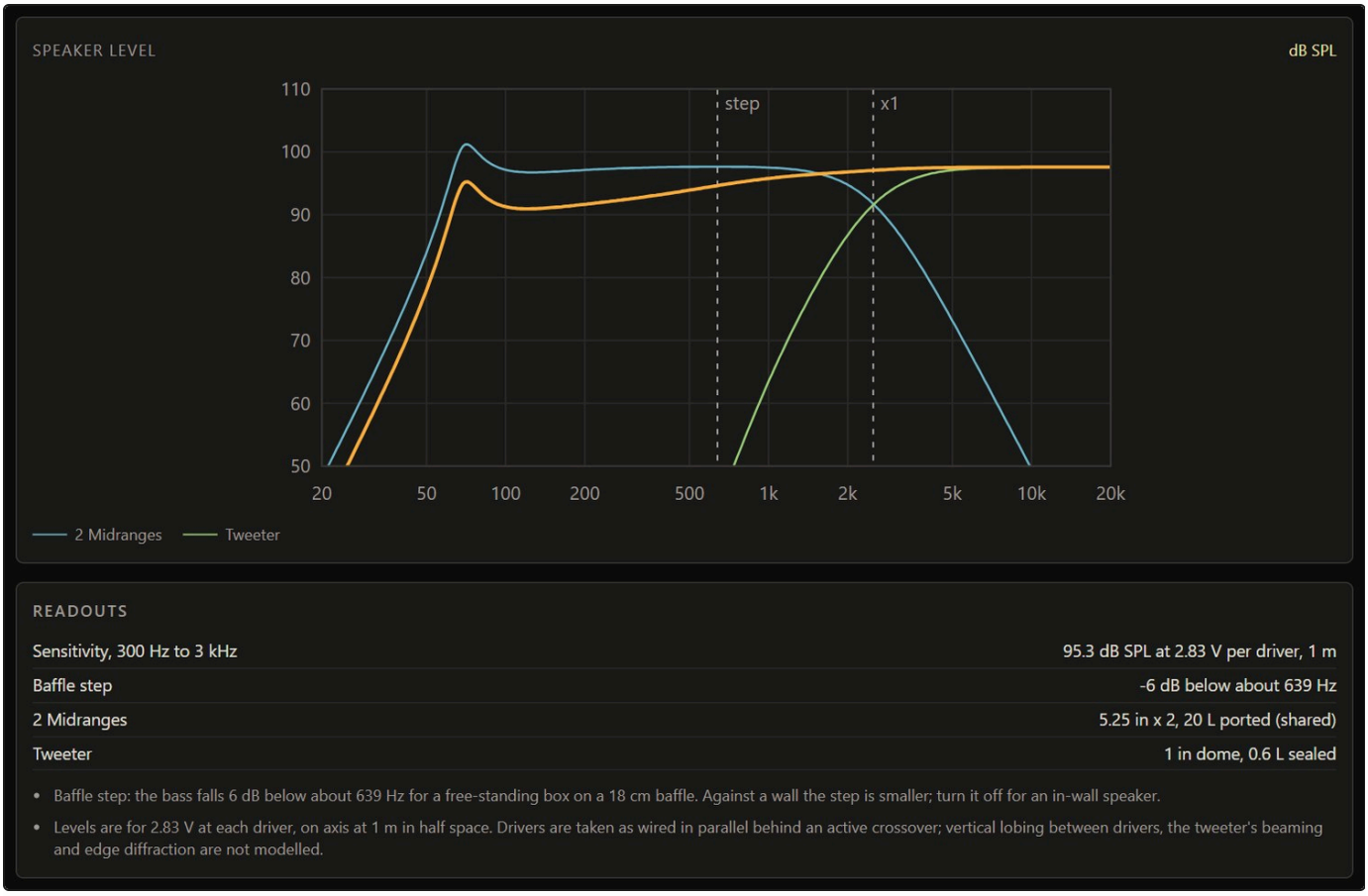
A driver's section

- **Driver.** Any class, or Custom with the eight datasheet fields (chapter 15).
- **No box (open baffle).** For a planar or ribbon tweeter with nothing behind it: a second-order high-pass at its own resonance.
- **Box** (sealed or ported), **Air** (for a pair, the air for all of its drivers; each gets its share), and for a ported box the **Tuning** and **Port** diameter. Choosing Ported fills in a tuning and a port that fit: tuned at $Fs\,(Vas / Vb)^{0.31}$, a port half the cone across, narrowed until the tube is at least 2 cm long. Edit them freely; the Build page warns if a port does not fit.
- **Trim** (-20 to +6 dB), **Set back** (how far the voice coil sits behind the baffle plane, 0 to 30 cm, which delays that driver) and **Invert**.



A dome tweeter needs no special model. It is a small driver, and a small sealed chamber behind it, so it uses the same sealed-box response as a woofer. Its resonance matters: cross it at least twice above its resonance or the page warns.

The plot and readouts



The thick line is the whole speaker, with the baffle step, in dB SPL at 1 m on axis for 2.83 V at each driver. The thin lines are each driver through its crossover leg (before the baffle step), the dashed lines the crossover points and the baffle step. Below it: **Sensitivity** (the average from 300 Hz to 3 kHz), the baffle-step frequency, each driver and its chamber, then warnings and notes. Warnings:

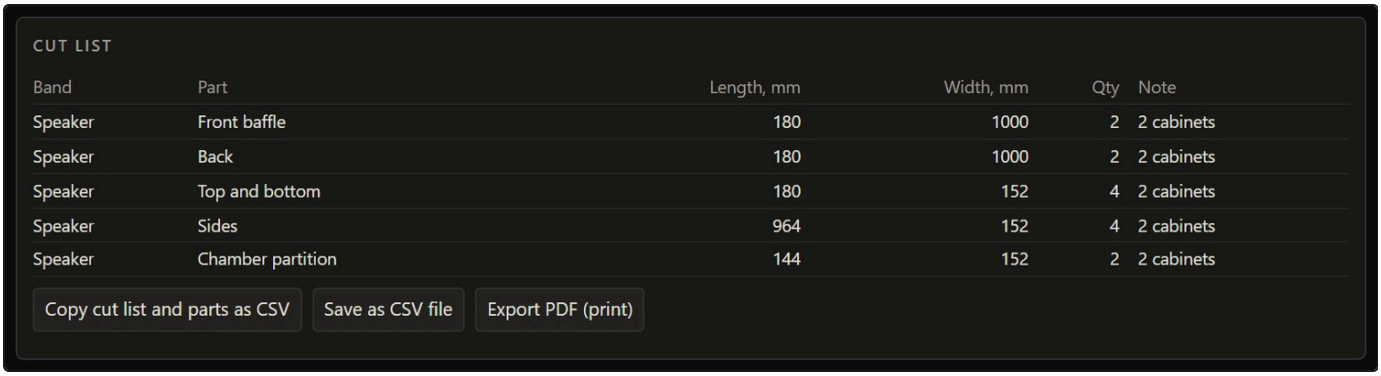
- a tweeter or midrange crossed under twice its resonance;
- a woofer crossed above where its cone starts to narrow its beam (the cone's circumference reaches about 3 wavelengths);
- a cone that does not fit the baffle width.



The 3-way floorstander: three drivers, two crossovers and the baffle step.

Building it

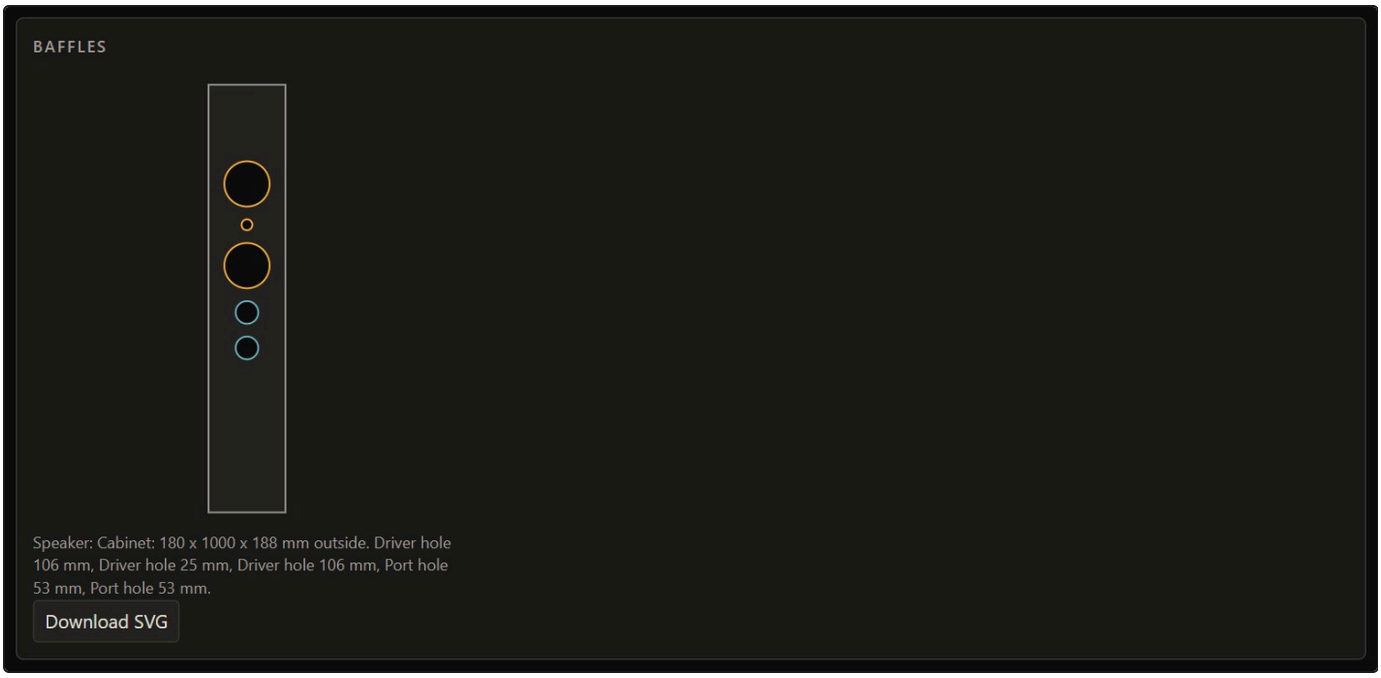
Press **Cut list and drawings** or open **Build**. The wall and sheet settings, units (millimetres or inches), the 3D preview, SVG and DXF files and the print layout work as in chapter 8; the per-band box shape section is replaced by the speaker's own sizes.



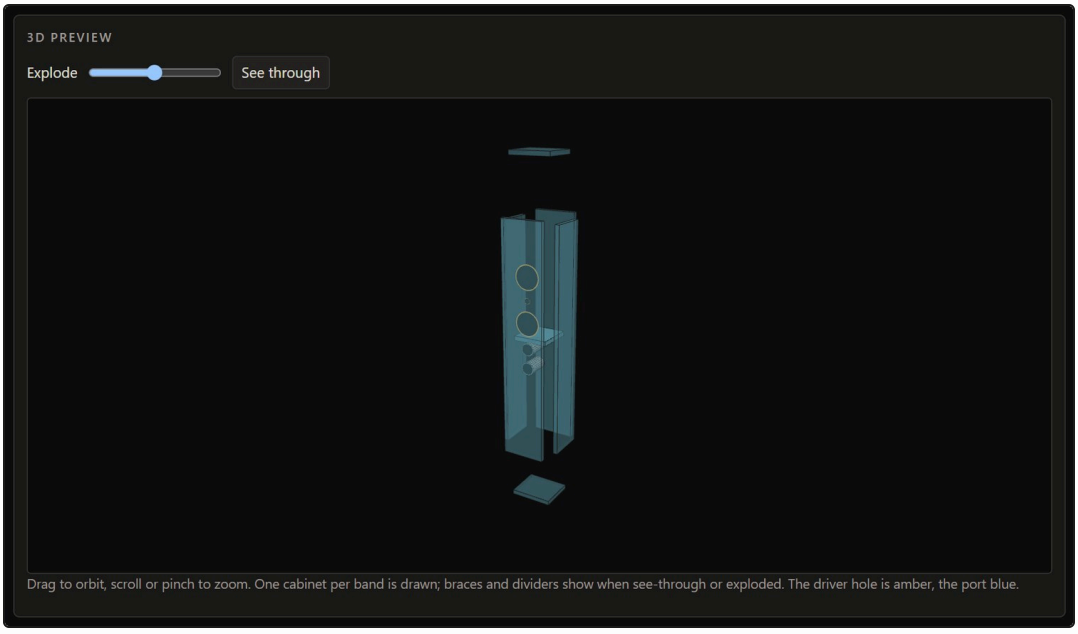
The MTM tower with separate woofer chambers: the one extra part is the chamber shelf.

- **One cabinet** at the baffle size you gave. Its depth follows from the air in the woofer and midrange chambers plus everything inside: the drivers, port tubes, chamber shelves and braces. A tweeter's air is not counted: a dome has its sealed back built into the driver.
- **Chamber shelves.** With separate chambers, each extra chamber adds one flat shelf (interior width by interior depth) between the drivers it divides. Its height is where the air above it runs out, so equal chambers put it halfway and a 3-way's shelf sits between the midrange and woofer chambers in proportion to their air. A shared chamber, a dome tweeter and an open-baffle driver add none. The 3D view draws the shelves (see through, or exploded).

- **Drivers on the baffle.** In the arrangement's order from the top (an MTM has the dome between two equal driver holes, a TMM the dome on top), centred across; the whole stack starts a third of the way down the spare height, and port holes go underneath the lowest driver. A ported pair gets one port for each driver.



The MTM tower: two driver holes either side of the dome, and the two port holes below.



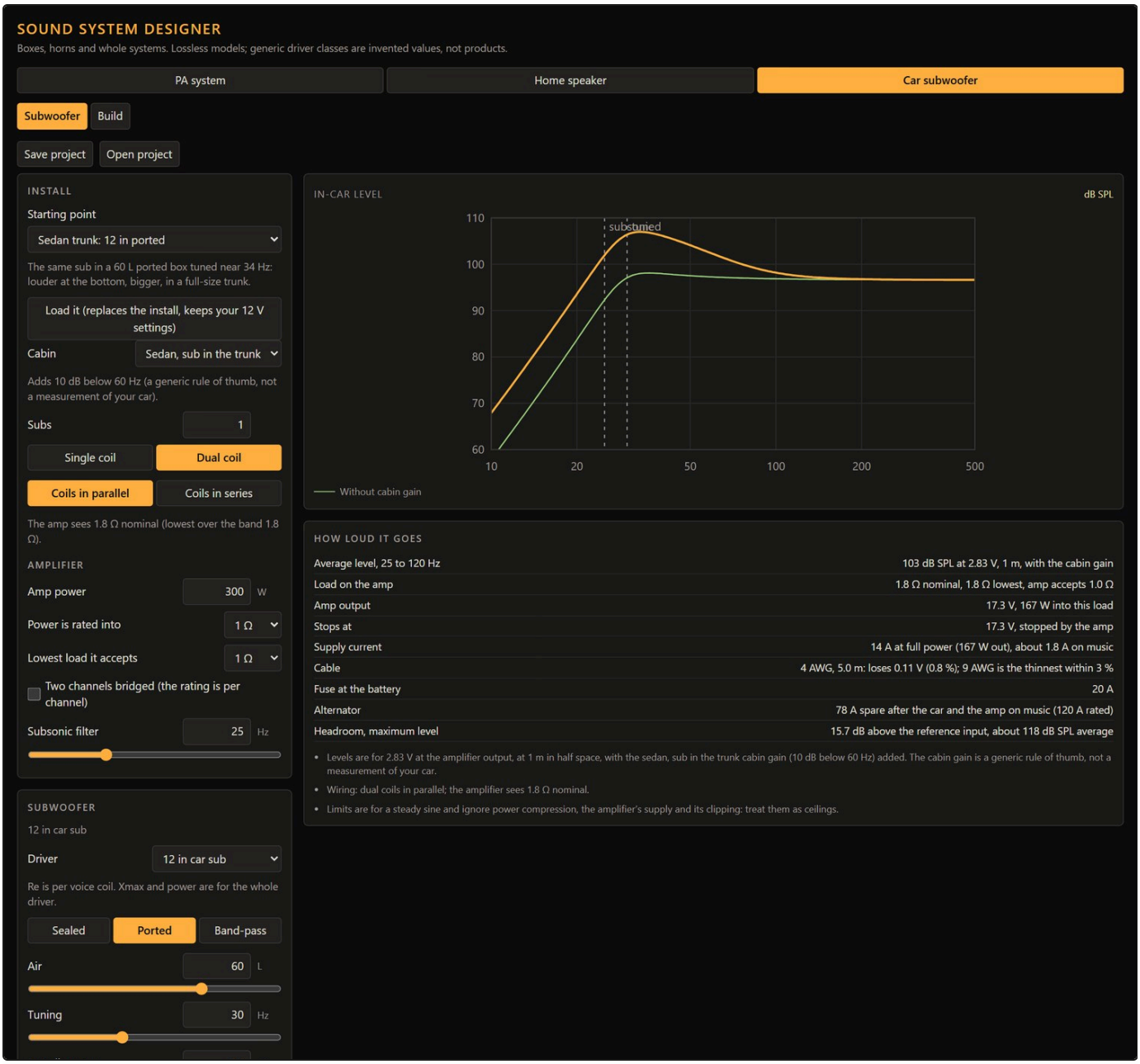
The 3D preview, exploded and see-through: every driver hole in amber, each port tube in blue, and the chamber shelf between the two woofer chambers.

The bill of materials counts every driver for every speaker you are building, one terminal per speaker, port tubes per port, screws, glue and damping. **The crossover is not in the list.** The speaker is designed with an active crossover; a passive network, its parts and the wiring are yours to add.

WHAT THE SPEAKER MODEL LEAVES OUT. On-axis response only: no vertical lobing between the drivers of an MTM, no tweeter beaming and no baffle-edge diffraction. Drivers are taken as wired in parallel behind an active crossover. A band-pass or horn slot is built as a sealed chamber (with a warning). The starting points use generic invented parts: confirm real driver parameters before you cut wood.

10 CAR AUDIO

Switch the header to **Car subwoofer** to design a car install: one or more subs in one enclosure, the amplifier and its wiring, the cabin they play into, the vehicle space the box has to fit, the 12 V supply, and optionally a pair of front speakers. The PA and home-speaker designs stay in the same project.



The Car page with the sedan trunk starting point.

Starting points

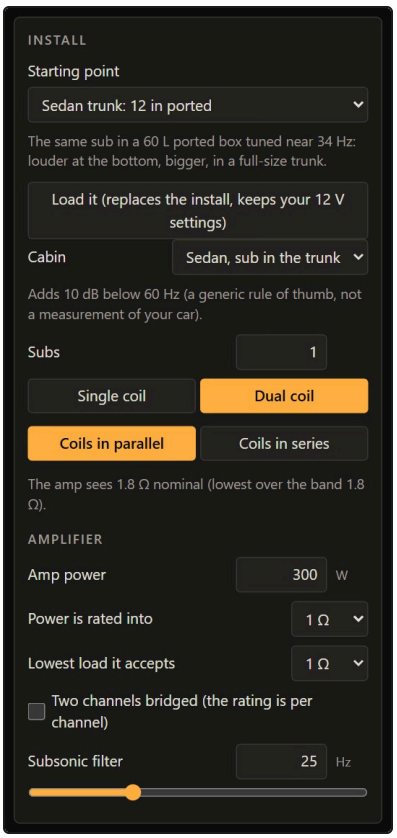
Choose one under **Starting point** and press **Load it (replaces the install, keeps your 12 V settings)**. Each is built to be clean: its amplifier just reaches the sub's own limit, its cable is the thinnest gauge that holds the voltage

drop within 3 %, and where it has front speakers they are levelled to the sub. Loading keeps your amplifier class, supply voltage, cable, alternator and the car's loads, and only thickens the cable if the new amp needs it.

STARTING POINT	WHAT IT IS	SPACE
Sedan trunk: 12 in sealed	12 in car sub, dual coil in parallel, 40 L sealed	`compact trunk`
Sedan trunk: 12 in ported	the same sub, 60 L ported near 34 Hz	`full-size trunk`
Sedan trunk: wedge against the seat	12 in sealed, 45 L, baffle sloping to follow the rear seat	`full-size trunk`
Hatchback: 10 in sealed	10 in car sub, 25 L sealed, small-cabin gain	`cargo floor`
SUV: two 10 in ported	two 10 in in one 70 L ported box, coils in series (3.6 Ω)	`cargo floor`
Under the seat: 10 in shallow	a shallow 10 in in a flat 10 L sealed box, firing up or down	`under a seat, flat`
Truck: two 10 in behind the seat	two 10 in in a wide, shallow 60 L ported box	`behind the seat`
Door speakers and a sub	the 12 in sealed sub plus a 6.5 in component set in the doors	`compact trunk`
Coaxials and a small sub	an 8 in sealed sub plus 6.5 in coaxials in the doors	`cargo floor`

These are generic, invented parts and sizes, not real products and not designs of any car. The amplifiers are sized to the sub's own sine-wave cone or heat limit, so they look small beside the driver's power rating: a steady sine at the worst frequency is a harsher test than music.

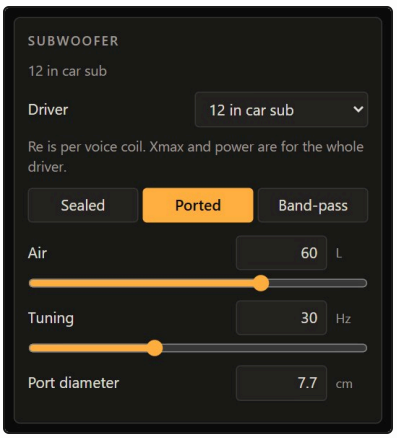
Install



- **Cabin.** The vehicle type sets the *cabin gain*: a small closed cabin pressurises at low frequencies, so a speaker in a car gets louder towards the bottom instead of falling away. Open (none), coupe or hatchback 12 dB below 70 Hz, sedan with the sub in the trunk 10 dB below 60 Hz, SUV or wagon 8 dB below 50 Hz, truck cab 9 dB below 65 Hz, van 6 dB below 40 Hz. These are rules of thumb: a real car varies by several dB with its leakage, seats and glass, so read the in-car curve as a trend.
- **Subs** (1 to 4) and how they are wired together when there is more than one (all in parallel, all in series, or strings).
- **Single or dual coil**, and for a dual coil whether its coils are in parallel or in series. A dual coil in series is as loud as a single coil at the same voltage; in parallel it is 6 dB louder and half the load. Four dual-3.6 Ω subs all in parallel are 0.45 Ω, which is why car amplifiers list minimums down to 0.5 Ω.
- **Amplifier:** power, the load it is rated into (0.5 to 8 Ω), the lowest load it accepts, and *two channels bridged* (the rating is then per channel; the swing doubles and so does the lowest load).
- **Subsonic filter** (10 to 60 Hz): the lowest frequency the install is asked to play; the cone's worst case is taken from there upward.

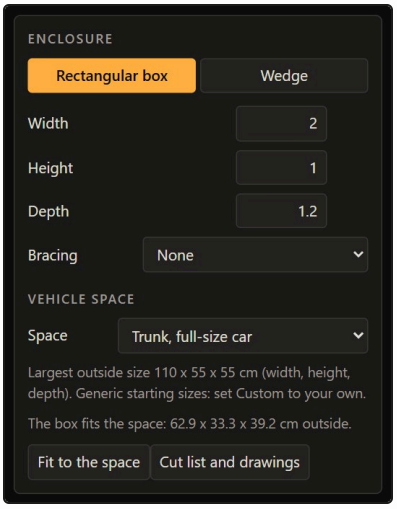
Subwoofer

- **Driver.** Six car classes (8, 10, 12 and 15 in, and shallow 10 and 12 in) come first in the list, then every other class and Custom. The generic values are invented. *Re is per voice coil; Xmax and power are for the whole driver.*
- **Box:** sealed, ported or band-pass; the air is for all the subs, each sub seeing its share. Choosing Ported fills in a tuning and a port that gives a tube about 20 cm long (a 12 in in 40 L: about 7 cm across, tuned near 34 Hz). A band-pass install is built as a box and a divider for each sub, the way a PA band-pass is.



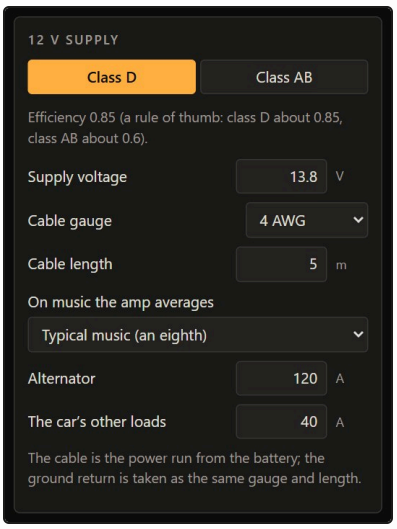
Enclosure and vehicle space

- **Rectangular box or wedge.** A wedge has a vertical back, flat top and bottom, and a baffle that slopes from the front of the bottom up to the front of the top: the shape that follows a rear seat. *Top depth / bottom depth* sets the slope; 1 is a plain box. A wedge has no bracing.
- **Proportions** of the inside (width : height : bottom depth) decide the shape; the app finds the size that holds the air you asked for after the drivers, ports and (for a box) braces are taken out.
- **Vehicle space.** Pick a generic space (compact or full-size trunk, hatchback or SUV cargo floor, under a seat flat or on edge, behind a truck seat) or Custom to type your own: its largest outside width, height and depth. A line says whether the box fits. Fit to the space sets the proportions to the space's inside, so the box fills it as far as the air allows.



12 V supply

- **Class D or class AB** (efficiency about 0.85 or 0.6), the **supply voltage** (13.8 V default), the **cable** (gauge from 0 to 12 AWG and length; the ground return is taken as the same gauge and length), how hard the amp is driven on **music** (a quarter, an eighth or a sixteenth of its rated power on average), the **alternator** rating and the **car's other loads**.
- The page reports the supply current at full power and on music, the volts the cable loses and the thinnest gauge that holds the loss within 3 % of the supply, the fuse at the battery (the next standard size over 1.25 times the current), and the alternator's spare current. A cable that loses too much, and an alternator that cannot keep up, each give a warning.

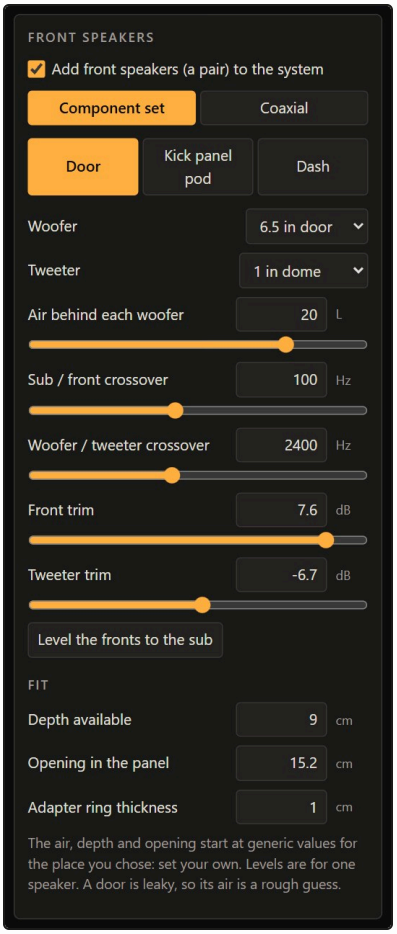


The check is a budget, not a simulation: no battery sag, no amplifier clipping from a collapsing rail, no cable inductance. The fuse is sized from the peak current and is not checked against the cable's own current rating: take that from the cable's maker.

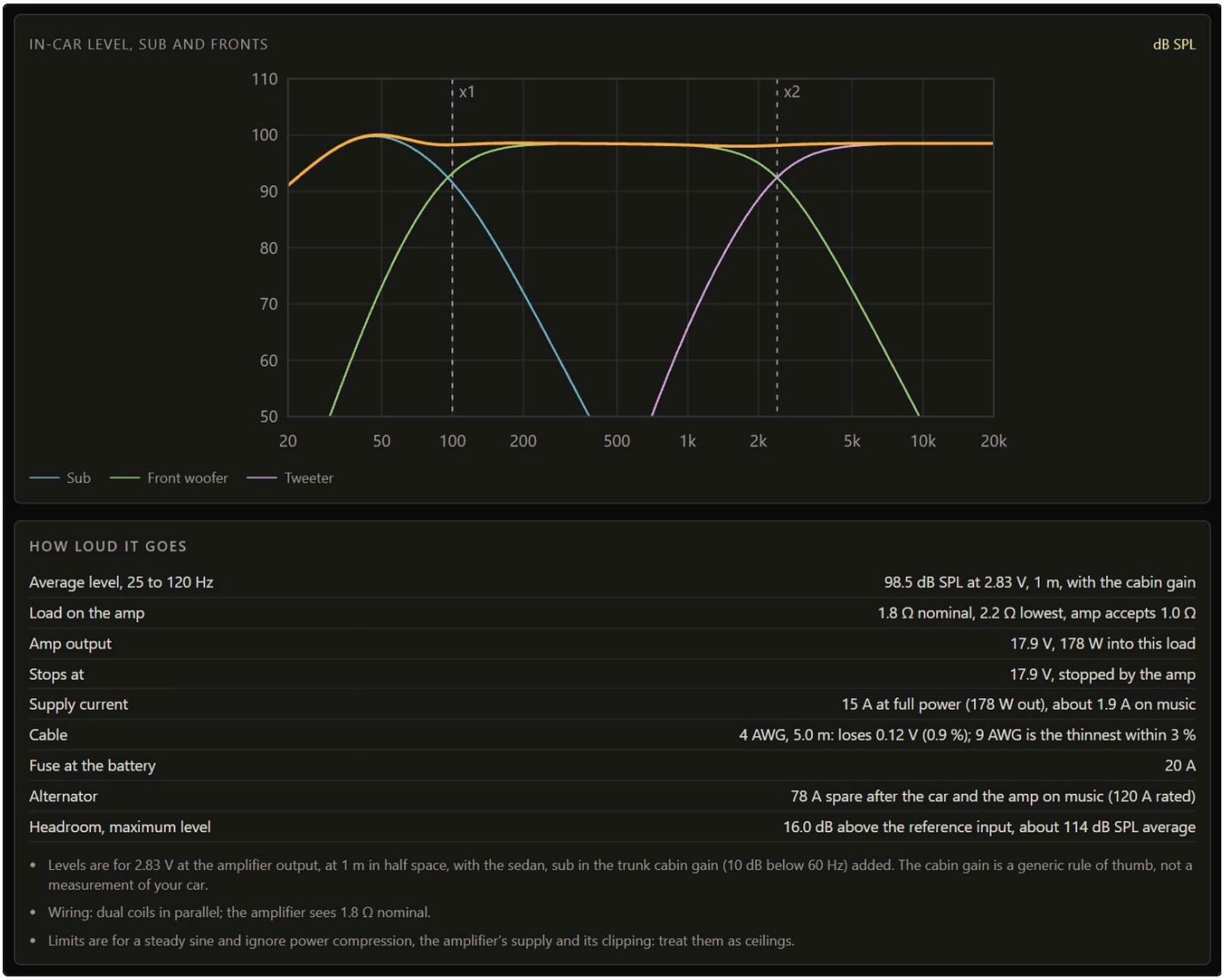
Front speakers

Tick **Add front speakers (a pair) to the system** to put a pair of door, kick panel or dash speakers in the plot with the sub.

- **Component set or coaxial** (a coaxial has its tweeter built in), the place (**door**, **kick panel pod** or **dash**, which start with that place's air and depth), the driver (5.25 in, 6.5 in or 6x9) and, for a component set, the tweeter (0.75, 1 or 1.25 in dome).
- **Air behind each woofer.** The front is modelled as a sealed box of the air you give it (20 L for a door). A real door leaks and has its own resonances, so this is a rough guess.
- **Crossovers:** sub to front (100 Hz) and woofer to tweeter (2.4 kHz), each a Linkwitz-Riley network; **trims** for the front and the tweeter.
  Level the fronts to the sub sets the front trim so the front woofer sits at the sub's level and the tweeter trim so the tweeter sits at the woofer's. Levels are for one speaker.
- **Fit:** the depth available behind the baffle, the opening in the panel and the adapter ring's thickness. Warnings: the depth is too small for the driver plus the ring, the opening is smaller than the driver's cut-out, the fronts are still rolling off at the sub crossover, a tweeter is crossed under twice its resonance, a woofer is crossed where its cone narrows its beam.



The plot and the limits



HOW LOUD IT GOES

Average level, 25 to 120 Hz	98.5 dB SPL at 2.83 V, 1 m, with the cabin gain
Load on the amp	1.8 Ω nominal, 2.2 Ω lowest, amp accepts 1.0 Ω
Amp output	17.9 V, 178 W into this load
Stops at	17.9 V, stopped by the amp
Supply current	15 A at full power (178 W out), about 1.9 A on music
Cable	4 AWG, 5.0 m: loses 0.12 V (0.9 %); 9 AWG is the thinnest within 3 %
Fuse at the battery	20 A
Alternator	78 A spare after the car and the amp on music (120 A rated)
Headroom, maximum level	16.0 dB above the reference input, about 114 dB SPL average

- Levels are for 2.83 V at the amplifier output, at 1 m in half space, with the sedan, sub in the trunk cabin gain (10 dB below 60 Hz) added. The cabin gain is a generic rule of thumb, not a measurement of your car.
- Wiring: dual coils in parallel; the amplifier sees 1.8 Ω nominal.
- Limits are for a steady sine and ignore power compression, the amplifier's supply and its clipping: treat them as ceilings.

The sedan door system: the sub, the front woofer and the tweeter, and the sum (thick) with the cabin gain.

With the fronts off the plot is the sub's in-car level, with a dashed line for the response without the cabin gain. With them on it is the whole system: the sub and the fronts through their crossovers, summed as complex pressure, with the cabin gain on the sum. **How loud it goes** gives the average level from 25 to 120 Hz, the load on the amp (nominal and lowest over the band, against what the amp accepts), the amp output, what stops the install first (the amp, the driver's power or the cone), the supply current, cable, fuse and alternator rows, and the headroom. Warnings: the load is under the amp's minimum; the amp can put more power into a driver than it is rated for; the cone passes Xmax.

Building it

Press Cut list and drawings or open Build. The wall and sheet settings, units, the 3D preview, SVG and DXF files and the print layout work as in chapter 8.

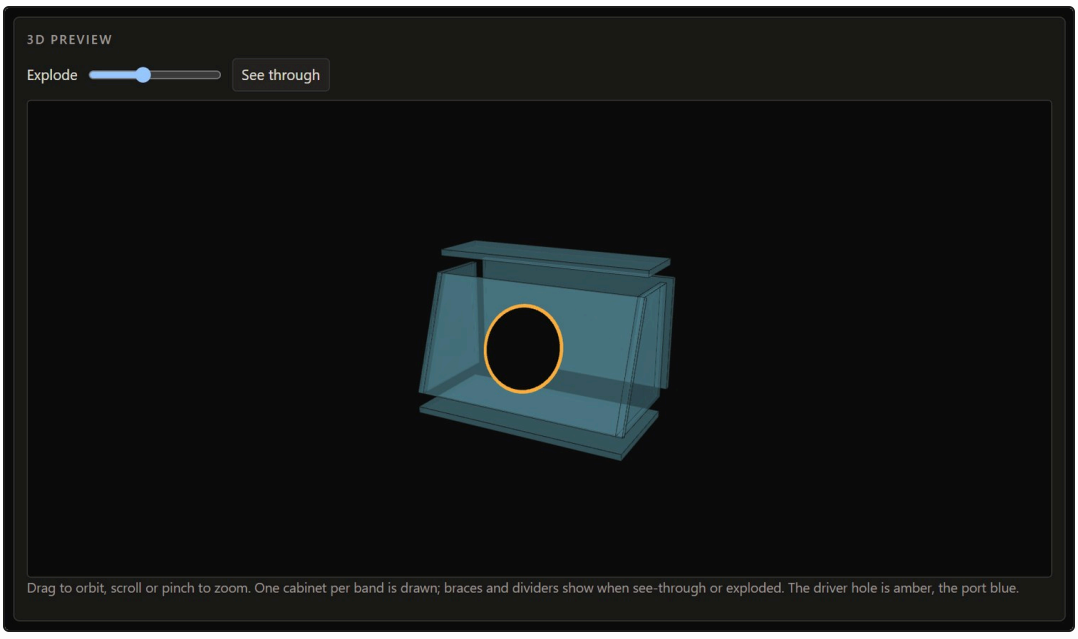
CUT LIST

Band	Part	Length, mm	Width, mm	Qty	Note
Subwoofer	Front baffle	549	321	1	
Subwoofer	Back	549	321	1	
Subwoofer	Top and bottom	549	285	2	
Subwoofer	Sides	285	285	2	
Subwoofer	6.5 in door adapter ring	182	182	2	1 cm thick; cut a round ring: the size listed is the outside's bounding rectangle, the cut-out is 14.2 cm round

Copy cut list and parts as CSV | Save as CSV file | Export PDF (print)

The cut list of the sedan door system: the sub box and the adapter rings.

- **One shared box** for all the subs (a sealed or ported box; a wedge has a back, a sloping baffle, a bottom, a top and two sides). The drivers sit in a row across the baffle, with the ports in the same row when they fit and in a row under them when they do not; a ported install has one port for each sub. The depth warns if the driver needs more behind the baffle than the box has (the top of a wedge is its shallowest part), and if a port is longer than the box is deep (fold it, narrow it, tune higher or add air).
- **Space fit.** The outside size is compared with the vehicle space; a wedge's depth counts as its bottom depth.
- **Adapter rings** for the front speakers: two, outside size the opening plus a 1.5 cm flange each side, the cut-out the driver's, four screw holes. The cut list gives the outside's bounding rectangle and the cut-out in the note; a drawing under Baffles shows the ring to scale (round, or an oval for a 6x9) with Download SVG.
- **Bill of materials:** the subs and the front speakers, the port tubes, a power cable and a ground cable of the gauge and length you set, a fuse and holder of the size the check gives, and the usual screws, glue and damping. The amplifier, ring terminals and the remote lead are not listed.



A wedge in the 3D preview, see-through and exploded: the sloping baffle with the driver hole lying on the slope.

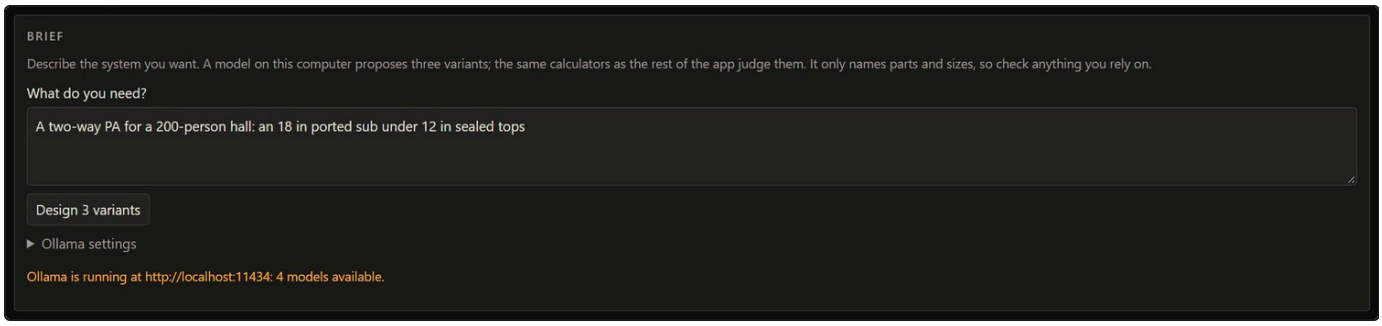


The baffle of the box and a front adapter ring.

WHAT THE CAR MODEL LEAVES OUT. The cabin gain is a generic shelf per vehicle type, not a measurement of any car, and cabin modes are not modelled. Limits are for a steady sine and ignore power compression and the supply: treat them as ceilings. A door is a sealed box of an air you guess. A wedge is cut with plain butt joints: expect to bevel the baffle edges and the sides. No battery sag, no time alignment between sub and fronts, no rear speakers, no passive crossover parts. Confirm real driver parameters and test the amplifier before you cut wood or wire a car.

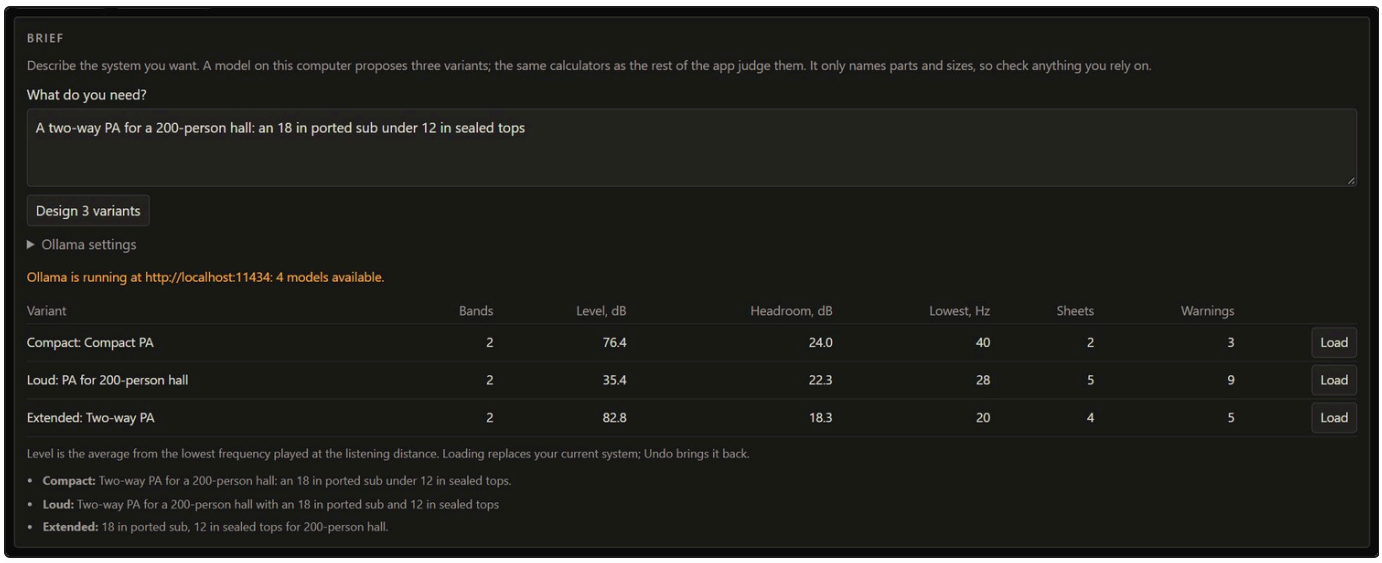
11 BRIEF: DESIGN FROM WORDS

Type what you want and a language model on your computer proposes three variants. Available on the System page (a whole system), on each box page (just that band's box), in Home speaker mode (one loudspeaker) and in Car subwoofer mode (a whole car install).



Using it

1. Open the page you want the design for. On a box page the Brief replaces *that band's box*; on the System page it replaces *the whole system*; in Home speaker mode it replaces the speaker, in Car subwoofer mode the car install (the PA system is kept).
2. Type a plain description, for example *A two-way PA for a 200-person hall: an 18 in ported sub under 12 in sealed tops*. Mention drivers, box types, number of ways, room size, or loudness. For a speaker: the arrangement (2-way, MTM, 3-way), the size and whether it is ported. For a car: the vehicle, the sub size and count, the power, and whether you want front speakers.
3. Press **Design 3 variants**. Expect roughly 5 seconds for a box and up to a minute for a system. **Stop** cancels.
4. Compare the table, then press **Load** on the one you want. It replaces your current design.
5. If you do not like it, press **Undo last load** to get your previous design back.

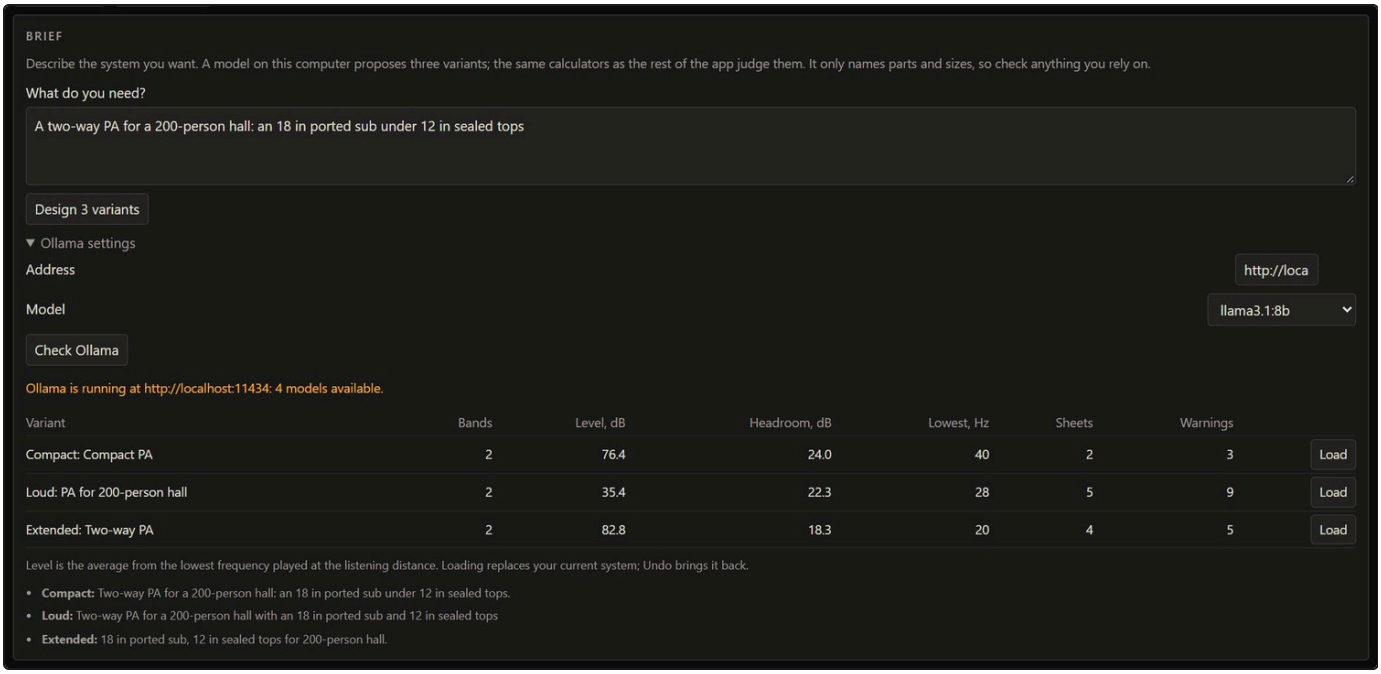


Three variants (Compact, Loud, Extended) compared. Warnings and the sheet count are the calculators' verdict.

Reading the table

For a system: the number of bands, the average level (dB at the lowest frequency played and listening distance), headroom, the lowest frequency, the sheet count and the number of warnings. For a box: the main readouts, volume, sheets and warnings. For a home speaker: the arrangement, the sensitivity (average level from 300 Hz to 3 kHz at the reference input, 1 m), the box's own -3 dB point before the baffle step, the total air, sheets and warnings. For a car install: the level in the car (25 to 120 Hz at the reference input), the headroom, the load the amplifier sees, the fuse size, sheets and warnings. A variant the model could not get right shows **Refused after N answers** with the reasons: the app checks every answer strictly (every number in range, the right type) and asks once more with the errors before giving up.

Ollama settings



- **Address:** where Ollama listens. Only `localhost` or `127.0.0.1` addresses are accepted; anything else is refused so nothing leaves your computer.
- **Model:** the list comes from Ollama. The tested default is `llama3.1:8b` ; other models may behave differently.
- Check Ollama asks the server and reports how many models it has. If Ollama is not running the app says so in plain words.

What the model chooses

For a **home speaker**: the arrangement, the baffle size, the crossover points, shared or separate chambers, the baffle step, how many to build, and for each driver group the driver class, box type, air, port tuning and port width. Only the generic home classes are offered (no horn driver, no custom driver); the program fills the trims by levelling the speaker like a starting point. For a **car install**: the cabin, the subs (class, count, wiring, box type and size), the amplifier (power, load, bridging, class, subsonic filter), the 12 V side (supply, cable gauge and length, music duty, alternator, the car's own load), the vehicle space and box shape, and optionally front speakers (type, mount, classes, air, depth, opening, ring, crossovers). The program levels the fronts to the sub. The fuse, the cable drop and the alternator margin are then judged by the calculators, not by the model.

> **JUDGE THE RESULT, NOT THE MODEL.** The model is not deterministic and sometimes gives a poor or refused design, especially for ported and band-pass boxes, multi-band systems and car installs with front speakers. It only names parts and sizes; the Warnings column is the calculators judging the result. Always review a loaded design on the System and box pages.

12 PROJECTS AND FILES

Saving and opening

- Save project opens a Save dialog and writes `<project-name>.ssd.json`. It holds the whole project: every band's box, the crossovers, trims, delays, drivers, amplifiers, the home speaker, the car install, shop settings and units.
- Open project reads one. The app checks the file: malformed, foreign, too-new or oversized files are refused with a message and nothing changes. A project you open replaces the current one; Undo last load brings the previous one back.
- The app also saves automatically on this computer and reopens your last project.

Exported files

WHAT	FROM	FORMAT	FILE NAME
Cut list and parts	Build	CSV	`<project>—cut-list.csv`
Baffle drawing	Build, Baffles	SVG	`<project>-<band>-<cabinet>—baffle.svg`
Sheet layout	Build, Sheets	SVG	`<project>—sheet-N.svg`
Sheet layout (CAD)	Build, Sheets	DXF (R12 ASCII)	`<project>—sheet-N.dxf`
Printable build sheet	Build	PDF via print	your choice
Project	header	JSON	`<project>.ssd.json`

The project name on the Build page becomes the start of every exported file name.

13 WHAT THE NUMBERS MEAN, AND THEIR LIMITS

How it calculates

Everything is evaluated in the frequency domain with the standard lumped-element models for sealed, ported and band-pass boxes (Thiele-Small), a transmission-line simulation for horns, and Linkwitz-Riley crossovers. The same formulas are shared with the Sound System VST3 plugin and are checked against published test vectors, closed-form results and the plugin itself.

Known limits

- **Horns:** one-dimensional and lossless, half space only. No bends, directivity or losses. The response is radiated power.
- **Systems:** active crossovers only. Identical drivers in a band are summed as one source (optimistic for wide arrays). Limits are small-signal: no power compression, no limiter. Each driver is taken to be in its

own box.

- **Home speakers:** active crossover only (no passive parts list); on-axis response (no vertical lobing, tweeter beaming or edge diffraction); a ported slot builds with front ports; a band-pass or horn slot is cut as sealed.
- **Car audio:** the cabin gain is a generic shelf per vehicle type; limits are for a steady sine (an amplifier matched to them looks small beside the driver's rating); front speakers sit in a sealed box of the air you give them; a wedge has no bracing; no battery sag, time alignment, rear speakers or passive crossover parts.
- **Build:** rectangular boxes with butt joints. Horn side panels are cut as bounding rectangles. Hole sizes and displacement are estimates. The sheet layout is a simple heuristic.
- **Drivers:** generic classes and values you enter. There is no driver database and no amplifier database.
- **Levels** are outdoor, half-space estimates, not a room simulation.
- **Brief:** a local language model is not deterministic and can be wrong. The calculators always have the last word.
- Windows only for now.

> **BEFORE YOU BUILD.** Use the plots to choose between designs, then confirm the real driver's measured parameters, check the amplifier's rating into your final load, and make a prototype or at least a mock-up. Loud systems can damage hearing and equipment.

14 TROUBLESHOOTING

PROBLEM	WHAT TO DO
Windows blocks the installer	SmartScreen warns about unsigned software: **More info**, then **Run anyway**.
The window is blank	Install or repair the Microsoft WebView2 runtime (included in Windows 11).
Design 3 variants is greyed out	Type something in the box, and make sure Ollama is running with a model installed. Open **Ollama settings** and press **Check Ollama**.
"Ollama is not reachable"	Start Ollama. The address must be `localhost` or `127.0.0.1` (default `http://localhost:11434`).
A variant says Refused	The model's answer did not pass the checks twice. Press Design again, shorten the brief, name the box type and driver size, or pick a different model.
Port "does not fit"	Use a smaller port diameter, a higher tuning, or a larger box.
Cone exceeds Xmax	Lower the amp power, raise the high-pass (the lowest frequency played), tune a ported box lower, or use a driver with more Xmax.
A dip at a crossover	Try polarity **Invert** on one band, a **Delay** on the earlier band, or move the crossover. A warning names bands that cancel.
A band cannot play as high or low as you want	Check its box and driver on the box page: a horn's flare cutoff limits its low end, and a compression driver limits the top.
A piece is bigger than a sheet	Change the box proportions, use larger sheets, or split the design.
Open project says the file is refused	Only files saved by this app (`.ssd.json`) from this or an older version open: a file saved by 1.2 or 1.3 does not open in 1.1 (the file version is 2), but 1.1 files open in 1.3; a 1.3 file with a car install opens in 1.2 as a PA system and the car install is ignored. Check it was not edited by hand.
The amp load is under its minimum	On the Car page, wire the coils in series, put some of the subs in series, or choose an amplifier that accepts a lower load.
The cable loses too much, or the alternator is short	Use the gauge the page names, shorten the run, or raise the alternator rating (the amp draws about a fifth to a quarter of its power on average on loud music).
The box does not fit the vehicle space	Press **Fit to the space**, reduce the air, choose a wedge, or use fewer subs.
The port is longer than the box is deep	Choose Ported again (it sizes a port for a 20 cm tube), or use a narrower port, tune higher or add air.
A speaker is much louder in the treble, or the tweeter looks too loud	Press **Level the drivers**. The classes are invented values, so a real driver's sensitivity will differ: enter it as Custom.
The speaker's port "does not fit"	Choose Ported again on that driver (it fills in a port that fits), or use a narrower port, a higher tuning or more air.

I lost my design	The app autosaves on this computer; reopen it. After a Brief load, **Undo last load** restores the previous design.

15 REFERENCE TABLES

Driver classes

CLASS	FS HZ	QES	QMS	VAS L	SD CM²	RE Ω	XMAX MM	POWER W
10 in	45	0.55	5	25	330	3.4	6	200
12 in	38	0.50	5	60	480	3.4	8	400
15 in	32	0.45	6	140	855	3.4	10	600
18 in	28	0.40	6	280	1230	3.4	12	900
21 in	24	0.38	6	450	1700	3.4	14	1200
8 in	65	0.50	4	18	215	6.0	4	150
6.5 in	85	0.55	4	8	140	6.0	3	100
1 in HF	700	0.30	3	0.002	8	6.0	0.5	40
3 in	140	0.70	3.5	0.7	33	4.0	2	20
4 in	115	0.60	3.5	1.8	55	4.0	2.5	40
5.25 in	100	0.55	3.5	3.5	88	4.0	3	60
3 in mid	180	0.70	3	0.6	33	6.0	1.5	40
4 in mid	150	0.65	3	1.5	55	6.0	2	50
0.75 in dome	1400	1.00	3	0.0075	2.2	4.0	0.4	20
1 in dome	900	1.00	2.5	0.025	5	4.0	0.5	40
1.25 in dome	700	0.90	2.5	0.05	8	6.0	0.6	50
Planar tweeter	1500	1.00	2.3	0.006	6	4.0	0.5	30
8 in car sub	45	0.60	5	15	220	3.6	9	300
10 in car sub	38	0.55	5	35	335	3.6	12	500
12 in car sub	32	0.50	5	50	490	3.6	14	800
15 in car sub	28	0.45	5	120	850	3.6	16	1200
10 in shallow sub	42	0.65	4	20	335	3.6	13	400
12 in shallow sub	36	0.60	4	35	490	3.6	15	600
5.25 in door	70	0.80	3.5	6	88	3.6	5	80
6.5 in door	60	0.75	3.5	12	140	3.6	6	100
6x9 door	55	0.70	3.5	20	210	3.6	7	120
5.25 in coax	75	0.85	3.5	6	88	3.6	5	70

6.5 in coax	65	0.80	3.5	12	140	3.6	6	90
6x9 coax	60	0.75	3.5	20	210	3.6	7	110

These are invented generic values, not real products. The first eight (10 in to 1 in HF) are the classes of the Sound System plugin; the nine after them are the home-speaker classes and the twelve after those the car classes (six subs, three door woofers and three coaxials; a coaxial's tweeter is built in). The 3 in, 4 in and 5.25 in are woofers; the dome and planar classes are tweeters, with a small Vas so that a tweeter is a few dB more efficient than a woofer, as real ones are.

Custom driver ranges

PARAMETER	RANGE	MEANING
Fs	`10 to 300 Hz`	Free-air resonance.
Qts	`0.15 to 1.5`	Total Q.
Qms	`1 to 20`	Mechanical Q.
Vas	`1 to 2000 L`	Equivalent compliance volume.
Sd	`5 to 2500 cm²`	Cone area.
Re	`1 to 32 Ω`	DC resistance.
Wedge	An enclosure whose baffle slopes, so it fits a trunk corner against a seat back.	
TMM / WTW	Tweeter, midrange, midrange / woofer, tweeter, woofer: the other orders of two drivers and a dome.	
Xmax	`0.2 to 40 mm`	Linear peak excursion.
Power	`5 to 5000 W`	Rated power.

Home speaker ranges

CONTROL	RANGE
Baffle width	`8 to 80 cm`
Baffle height	`10 to 200 cm`
Crossover	`100 Hz to 8 kHz`
Trim	`-20 to +6 dB`
Set back	`0 to 30 cm`
Speakers to build	`1 to 8`
Box air, tuning, port	`as the box page`

Car audio ranges

CONTROL	RANGE
Subs	1 to 4
Amp power	50 to 20000 W
Amp rated into / lowest load	0.5, 1, 2, 4, 8 Ω / 0.5, 1, 2, 4 Ω
Subsonic filter	10 to 60 Hz
Wedge top depth / bottom depth	0.15 to 1
Supply voltage	10 to 15.5 V
Cable gauge / length	0, 2, 4, 6, 8, 10, 12 AWG / 0.5 to 15 m
Alternator / the car's other loads	30 to 400 A / 0 to 200 A
Front: air behind each woofer	0.5 to 60 L
Front: depth available / opening / ring	2 to 30 cm / 8 to 30 cm / 0 to 5 cm
Front: sub crossover / tweeter crossover	50 to 250 Hz / 1 to 8 kHz
Front trim / tweeter trim	-30 to +12 dB / -20 to +6 dB

Control ranges

CONTROL	RANGE
Box volume	0.1 to 1000 L
Tuning	10 to 200 Hz
Front volume (band-pass)	1 to 1000 L
Port diameter	2 to 40 cm
Amp power (box page)	1 to 5000 W
Flare cutoff	15 to 2000 Hz
T (hyperbolic)	0 to 1
Throat area	1 to 5000 cm²
Mouth area	2 to 60000 cm²
Length (conical)	0.05 to 6 m
Throat chamber	0 to 50 L
Horn width	5 to 300 cm
Crossover	20 Hz to 16 kHz
Lowest frequency played	20 to 200 Hz
Listening distance	1 to 200 m
Trim	-40 to +12 dB
Delay	0 to 20 ms
Drivers per band	1 to 16
Amp power (system)	1 to 20000 W
Amp minimum load	1 to 16 Ω
Wall thickness	3 to 50 mm
Sheet length / width	300 to 6000 mm / 300 to 3000 mm
Saw kerf	0 to 10 mm
Box proportions W, H, D	0.2 to 5 each

16 GLOSSARY

TERM	MEANING
2.5-way	Two woofers and a tweeter, the second woofer playing only the bass.
Band	One slice of the frequency range, played by one box through its own amplifier.
AWG	American wire gauge: the lower the number, the thicker the cable.
Baffle	The front panel that carries the driver.
Baffle step	The 6 dB fall in bass of a free-standing box below the frequency where its baffle is about half a wavelength across.
Bridged	Two amplifier channels working together on one load: twice the swing, and a load of at least twice a channel's minimum.
Cabin gain	The rise in bass level of a speaker in a small closed cabin such as a car, compared with open air.
Coaxial	A woofer with a tweeter mounted on its axis, so one driver covers the whole range.
Component set	A separate woofer and tweeter, crossed over.
D'Appolito / MTM	Midrange, tweeter, midrange: a dome between two matched drivers.
Dual voice coil (DVC)	A driver with two voice coils on one cone, wired in series or parallel to set its load.
Dome	A small tweeter whose diaphragm is a dome.
F3	The frequency where the response is 3 dB below the passband.
fb / fc	Port tuning / box resonance or flare cutoff.
Flare	The way a horn's cross-section grows from throat to mouth.
LR4	Linkwitz-Riley fourth-order crossover; adjacent bands sum flat.
Mouth / throat	The large (open) and small (driver) ends of a horn.
Open baffle	A driver with nothing sealed behind it, as a planar or ribbon tweeter.
Subsonic filter	A high-pass at 20 Hz or so that keeps the amplifier from driving the cone with inaudible low frequencies.
Qtc	The total Q of a sealed box and driver; about 0.7 is a classic flat alignment.
Scoop	A rear-loaded bass horn: the cone's front radiates directly, its rear loads a horn.
SPL	Sound pressure level in dB.
Thiele-Small	The standard set of parameters (Fs, Qts, Vas and so on) that describe a driver.
Xmax	The cone's linear peak excursion; beyond it distortion rises quickly.

About this program

Sound System Designer 1.4.0 is free software under the GNU Affero General Public License, version 3 only. Source: `https://github.com/spinsycle/sound.system.designer`. Third-party components are listed in `THIRD-PARTY-NOTICES.md`. Generic parts only: no brand, driver model or named sound system is modelled or implied. It makes no claims about the safety or performance of anything you build.